



**GHENT
UNIVERSITY**

SUMMER SCHOOL AI4AS 2026

PITFALLS IN MACHINE LEARNING

Jeroen Degroote
Stijn Heirbaut

OVERVIEW

OVERVIEW TOPICS

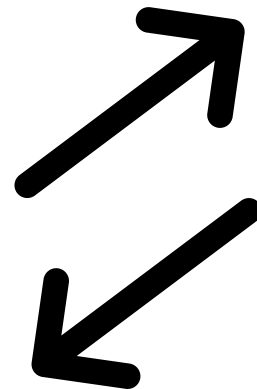
1. Cross-validation and overfitting
2. Data leakage
3. Grouped data dependence
4. Data imbalance
5. Feature dependence (multicollinearity)

SET-UP OF THE COURSE

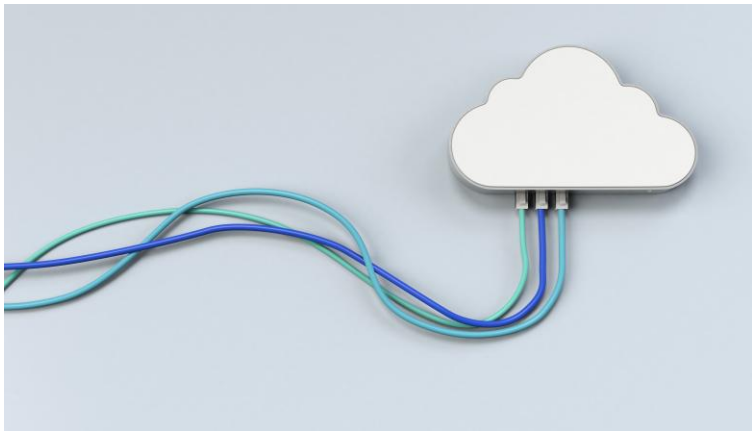
- Github:
 - ai4as_pitfalls_in_machine_learning
 - Presentation and code
- By topic(s)
 - Theory section: presentation
 - Practical exercise

GITHUB CODESPACE

Github repository



Github codespace



Local PC



+

Reproducible (cloud) environment

+

Easy set-up (dev container)

+

Github copilot

Account plan

Storage per month

Compute time per month

GitHub Free for personal accounts

15 GB-month

120 hrs

GitHub Pro

20 GB-month

180 hrs

Your space, your way. Codespaces is a home away from home for your code that feels just like your usual machine.

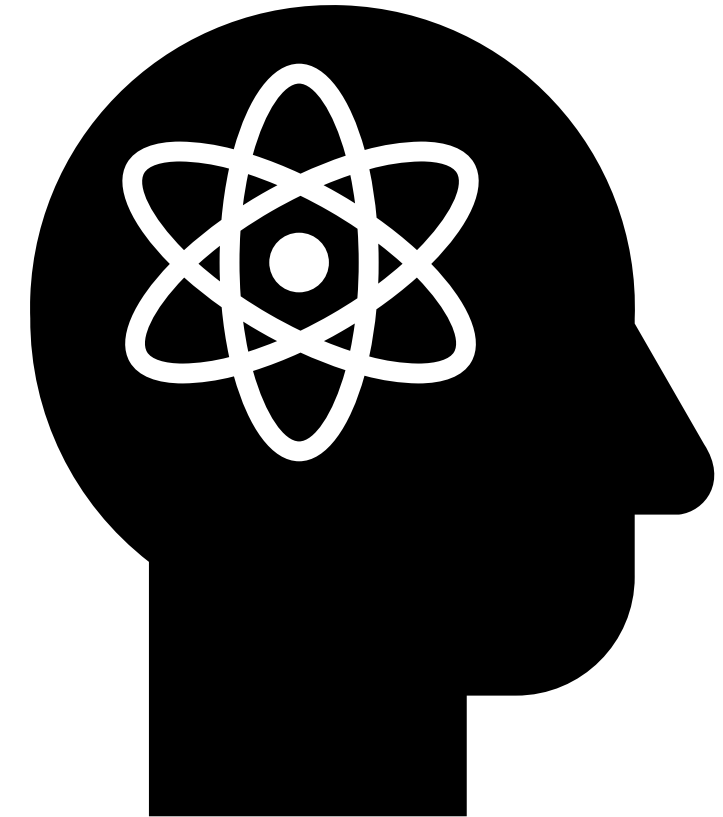
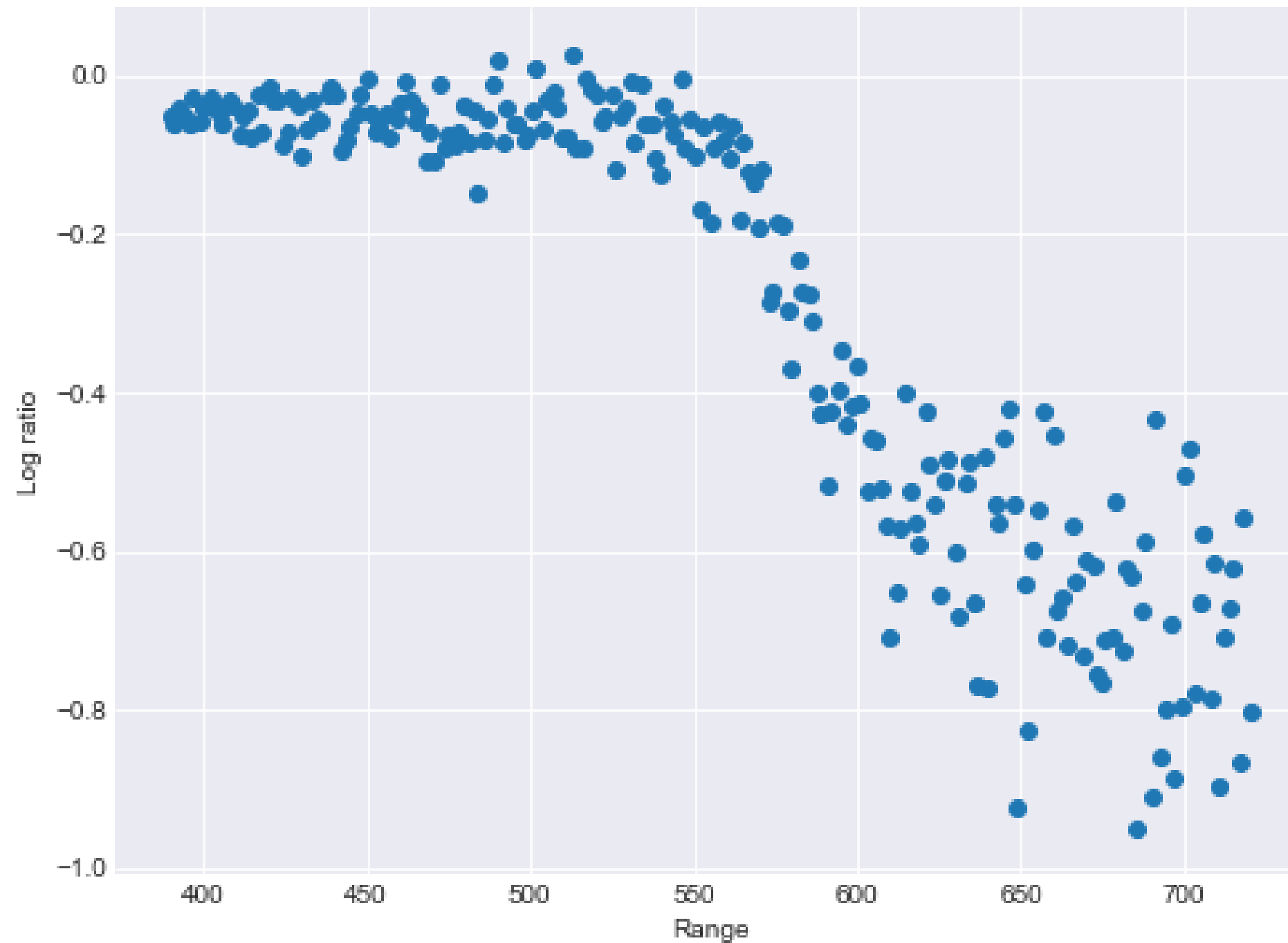
- Start coding instantly from anywhere in the world. Switching projects? Grab a new machine from the cloud that's preconfigured for that project. Your settings travel with you.
- Tabs or spaces? Monokai or Solarized? Prettier or Beautify? It's up to you. Control every nerdy detail only you care about with your own dotfiles repository.

GITHUB CODESPACE

- Fork *ai4as_pitfalls_in_machine_learning*
- Detailed instructions: see README
- Create codespace (recommended 16 GB)
 - Dev container builds automatically
 - Select Python kernel

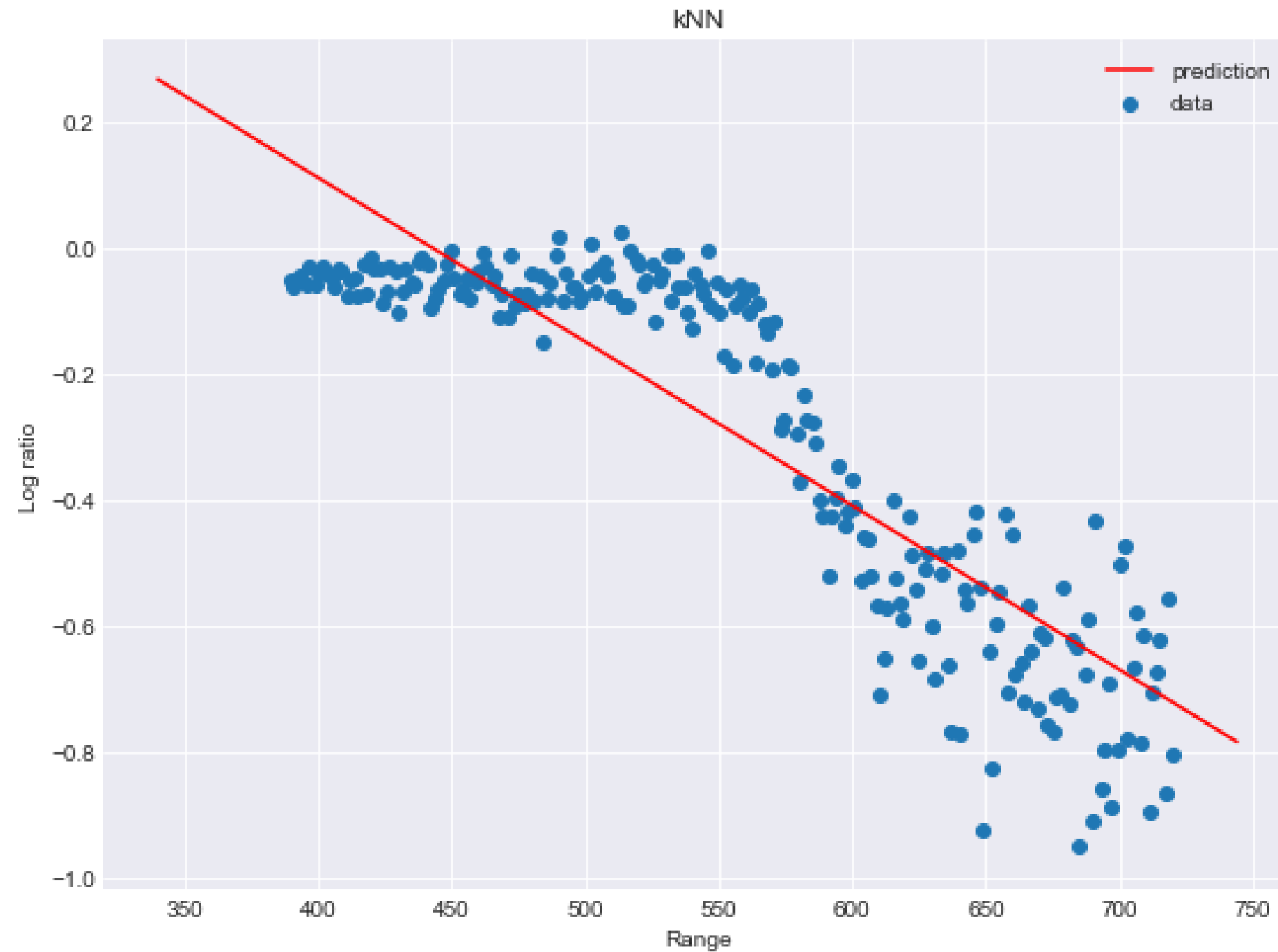
CROSS-VALIDATION AND OVERFITTING

INTRODUCTION: A MODELING TASK

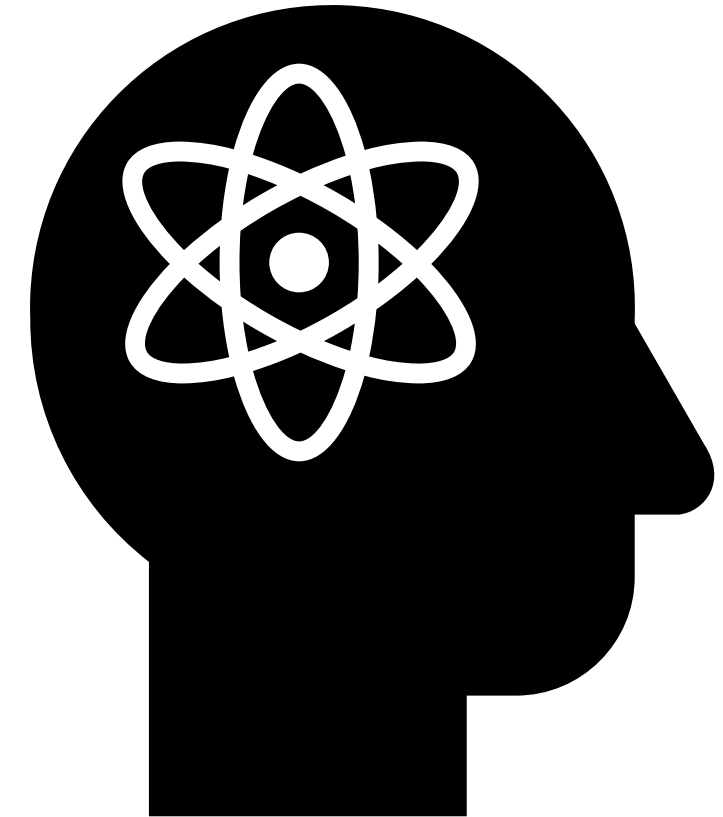


Which model?

INTRODUCTION: A MODELING TASK

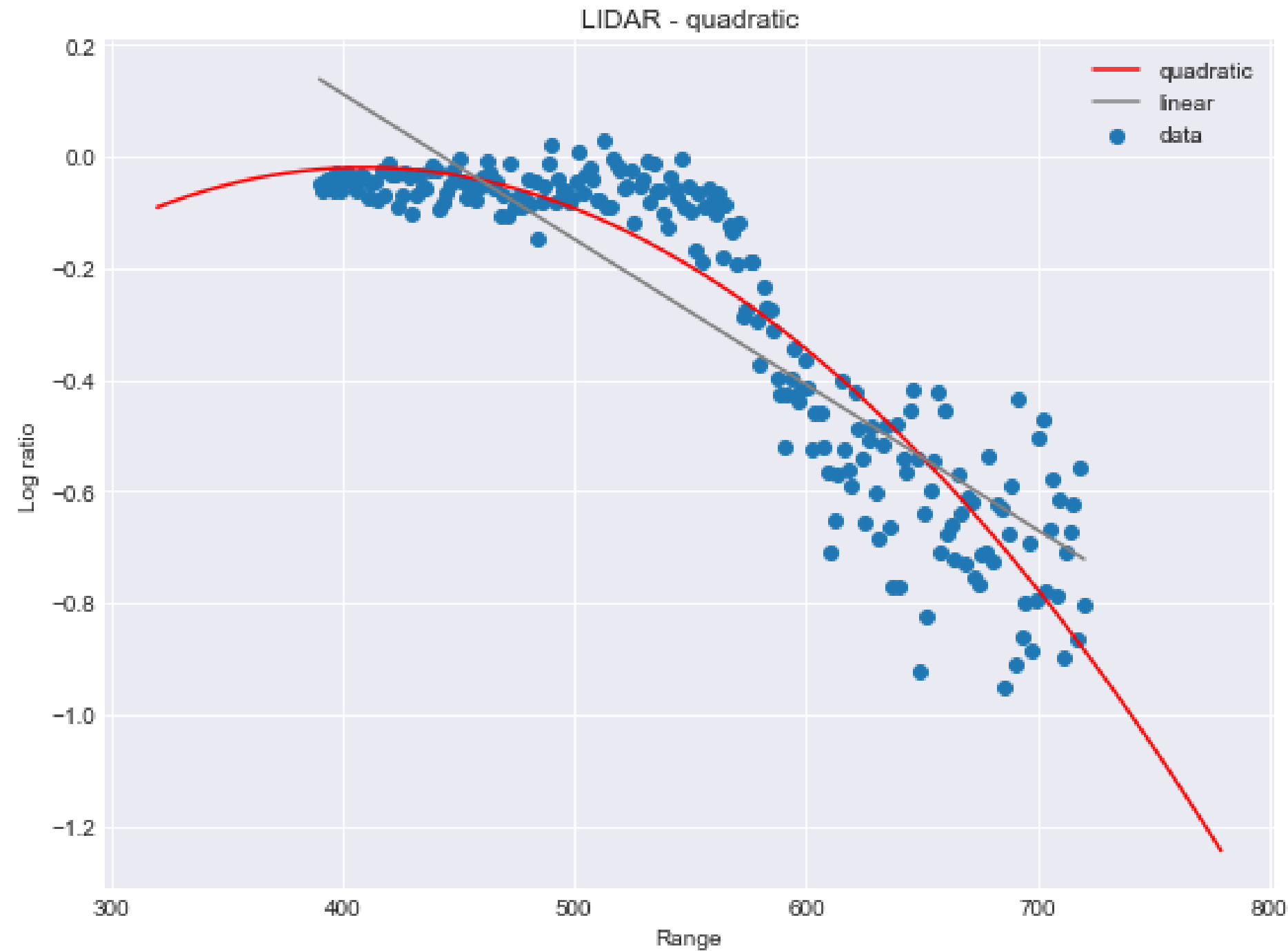


Improve?

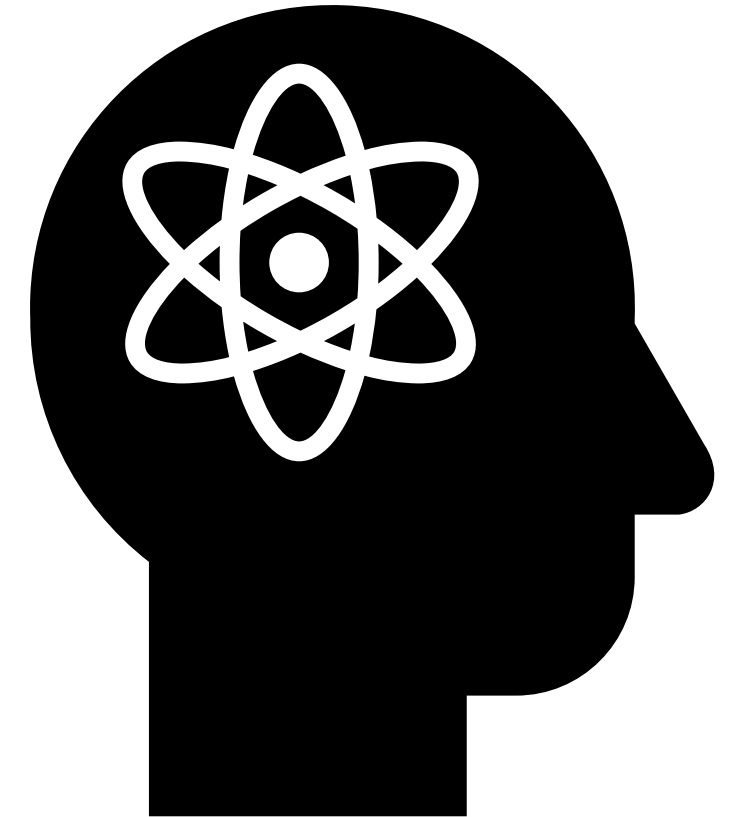


Linear regression

INTRODUCTION: A MODELING TASK

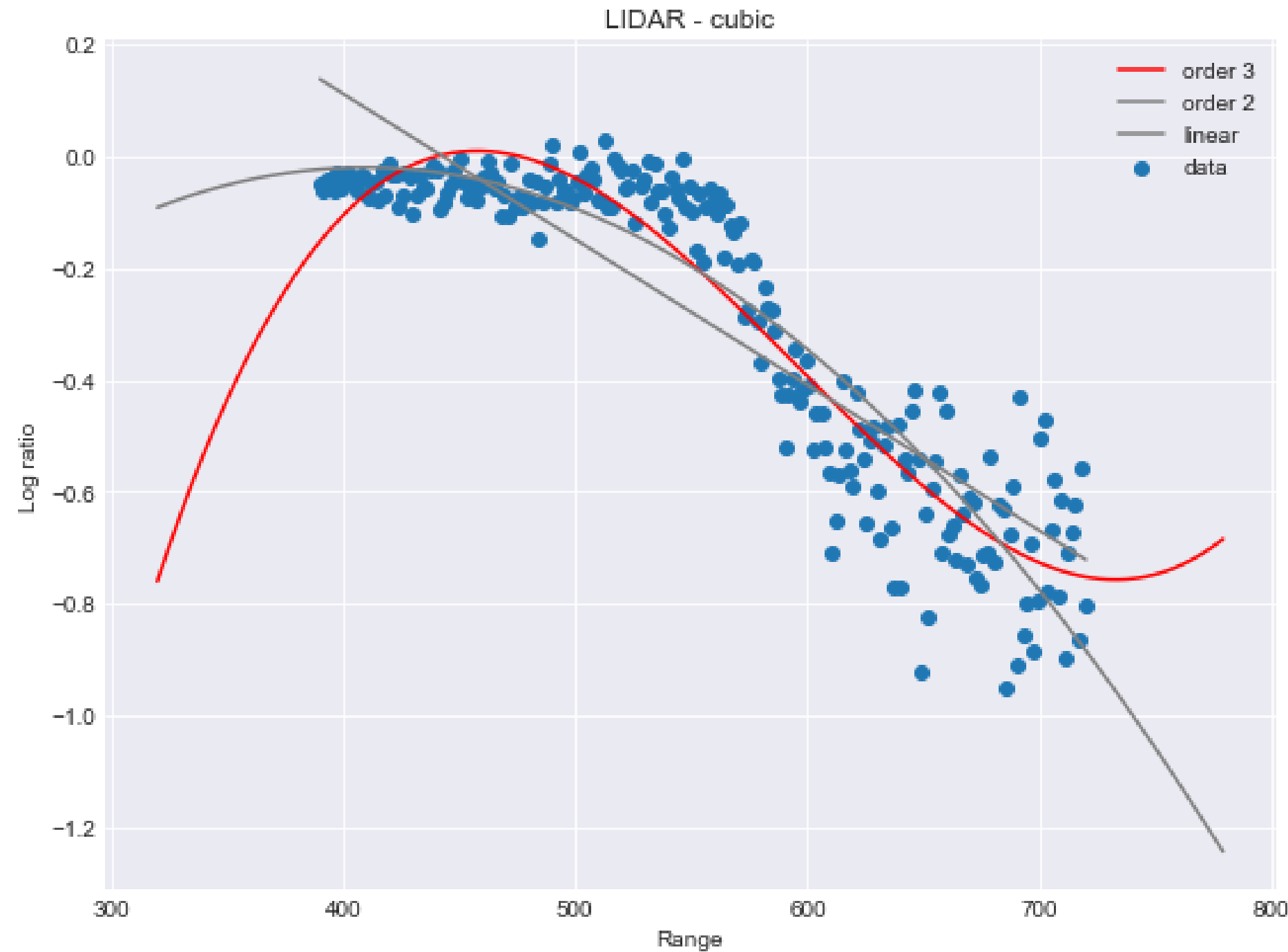


Improve more?

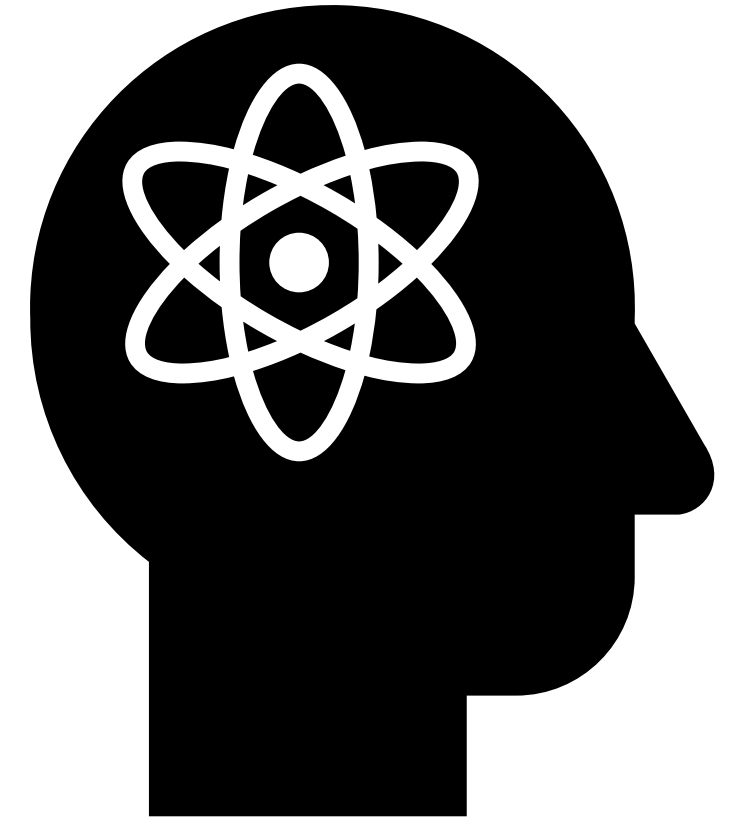


Quadratic polynomial

INTRODUCTION: A MODELING TASK



Improve more...?



Order 3 polynomial

INTRODUCTION: A MODELING TASK

- Problem: optimizing on 1 dataset = overoptimistic
 - Minimizing cost function
 - Tuning: selection order of model
 - Which model? → linear model, RF, NN

How is performance on new data?

Overfitting: well optimized on train data, low generalization

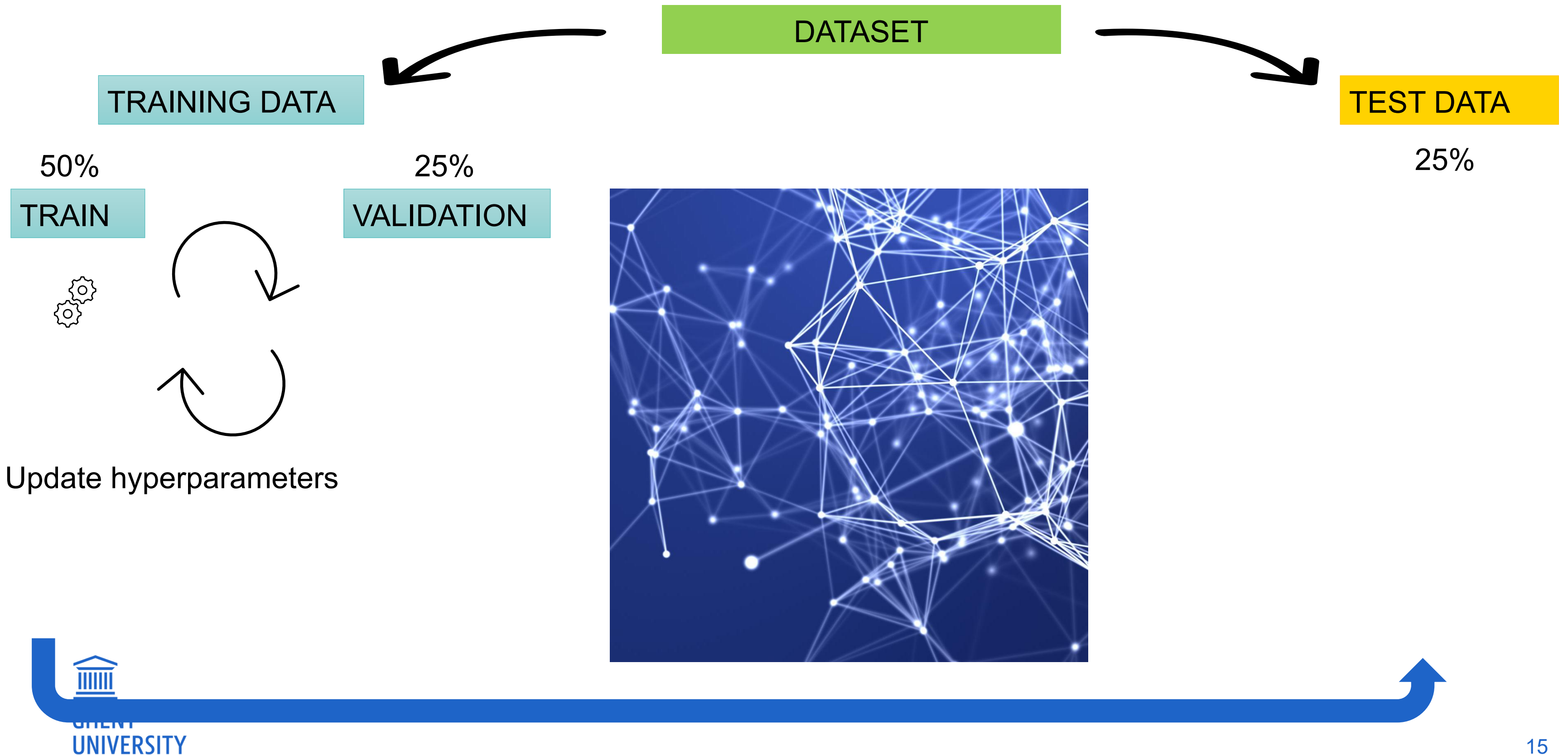
TRAIN – TEST DATA



How is performance on new data?

Overfitting: well optimized on train data, low generalization

TRAIN – TEST DATA: SCENARIO 1



TRAIN – TEST DATA: SCENARIO 2

TRAINING DATA

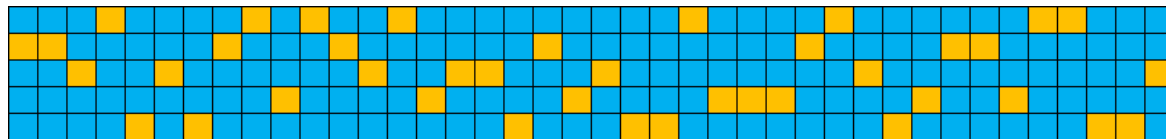
75%

DATASET

TEST DATA

25%

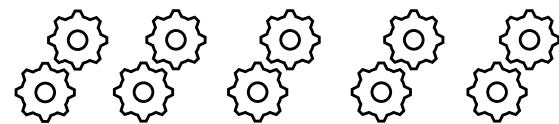
K-FOLD CROSS-VALIDATION



5 fold CV

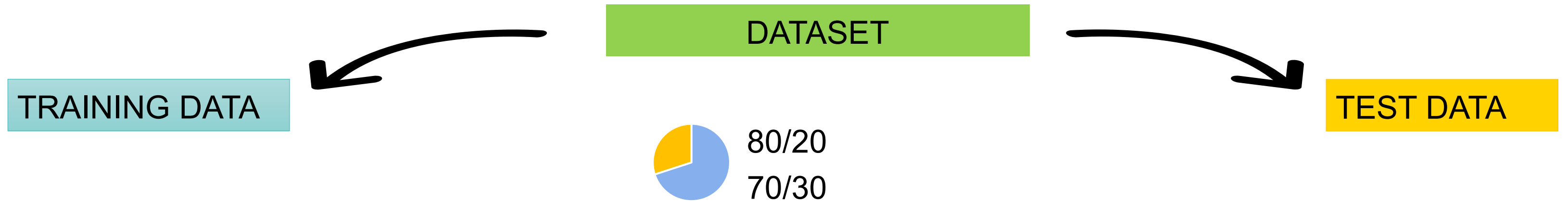
- 4 folds training + tuning
- 1 fold testing

For each hyperparameter

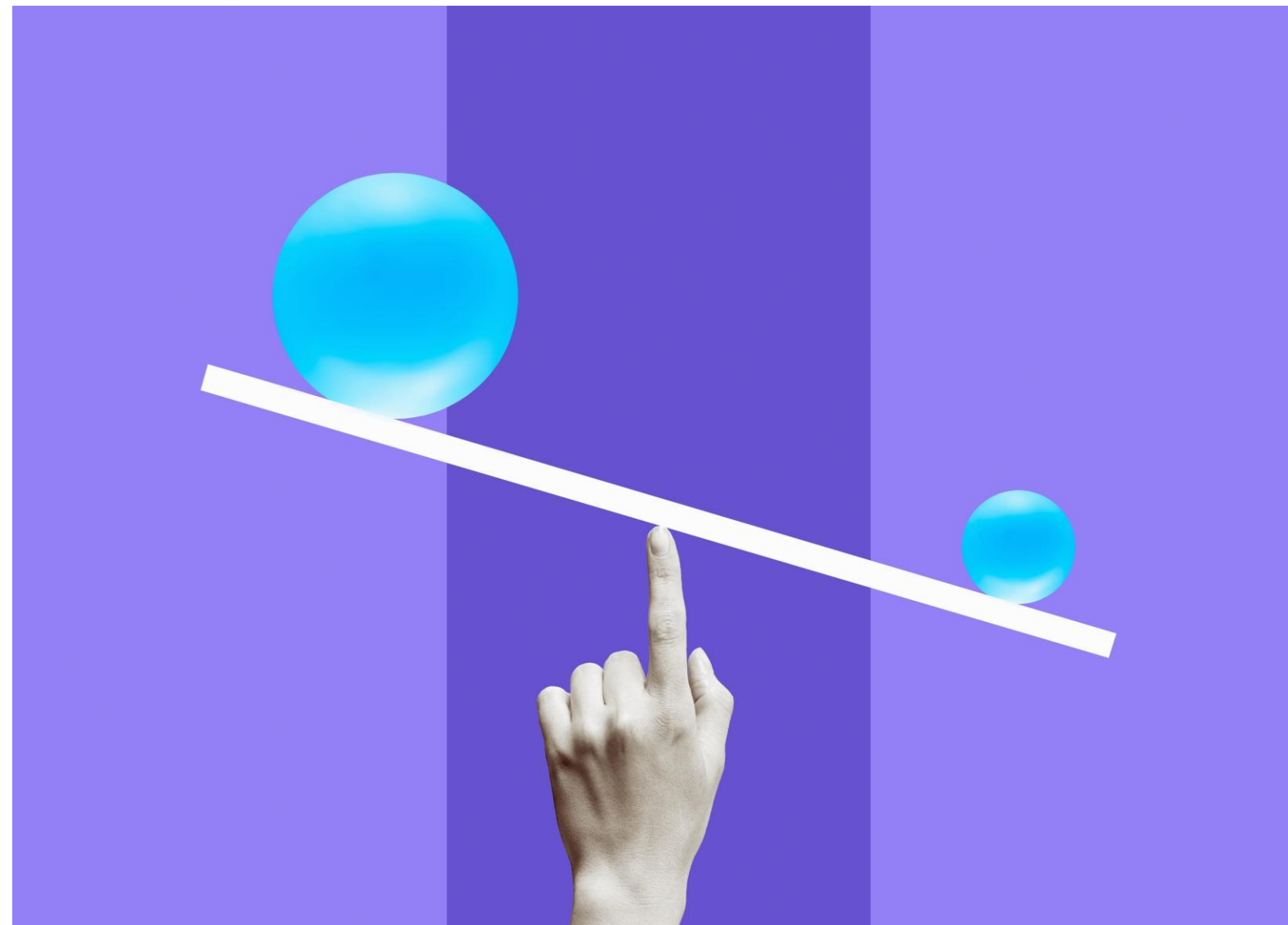


Refit TRAIN: best hyperparameter

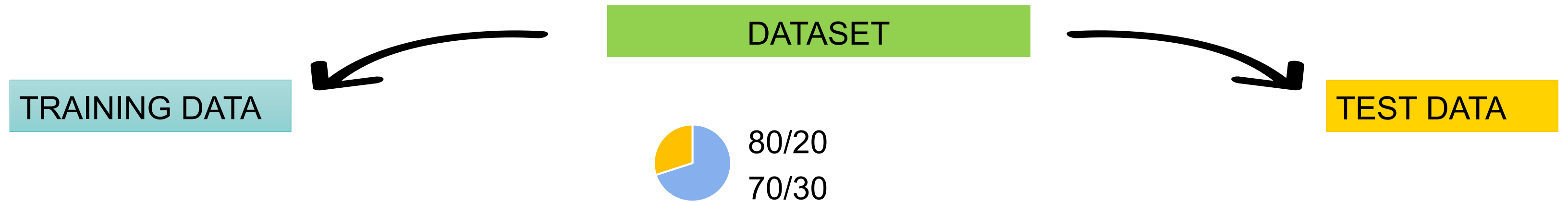
TRAIN – TEST DATA: SCENARIO 3



Limited data (animal science)

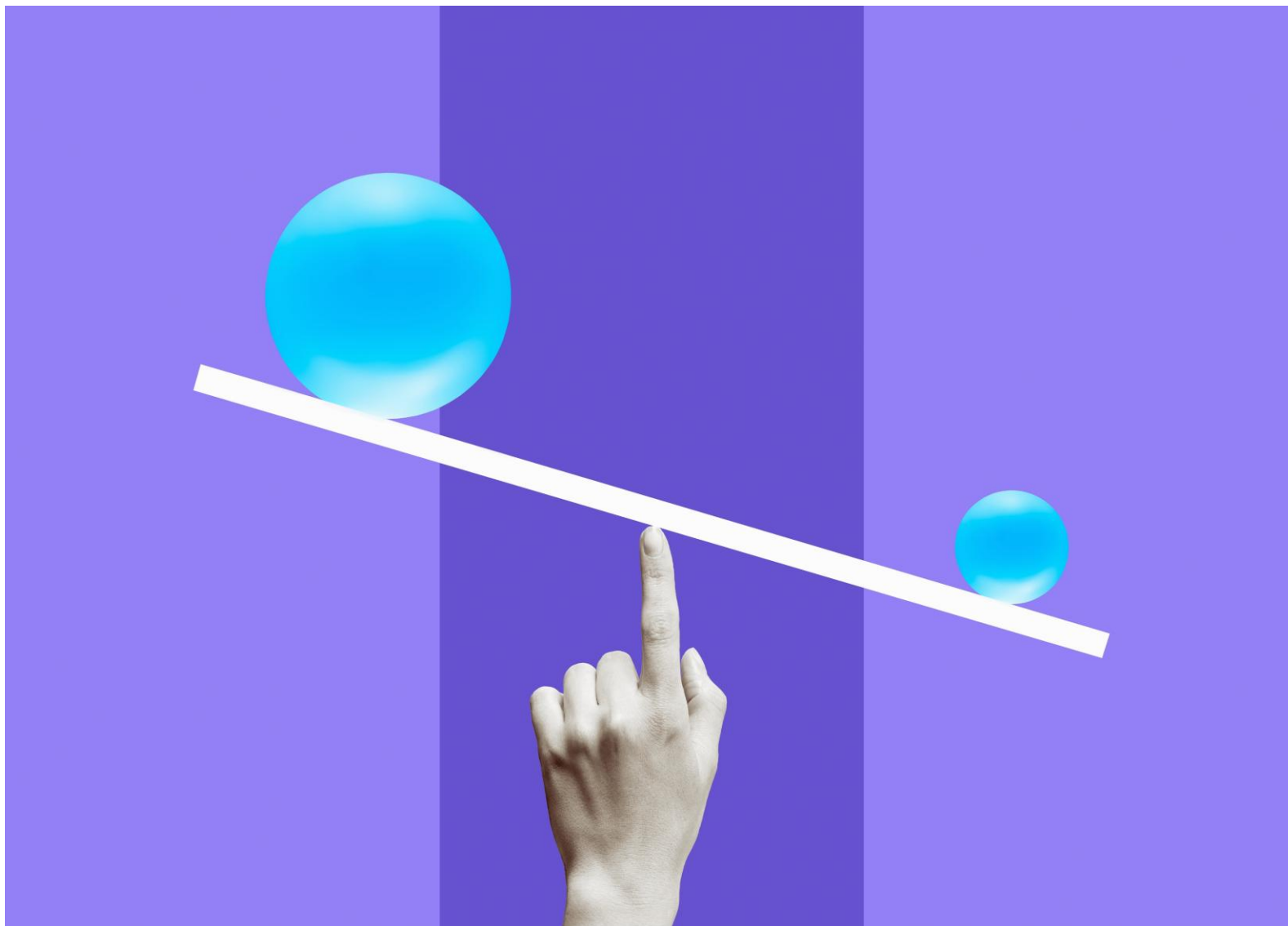


TRAIN – TEST DATA: SCENARIO 3

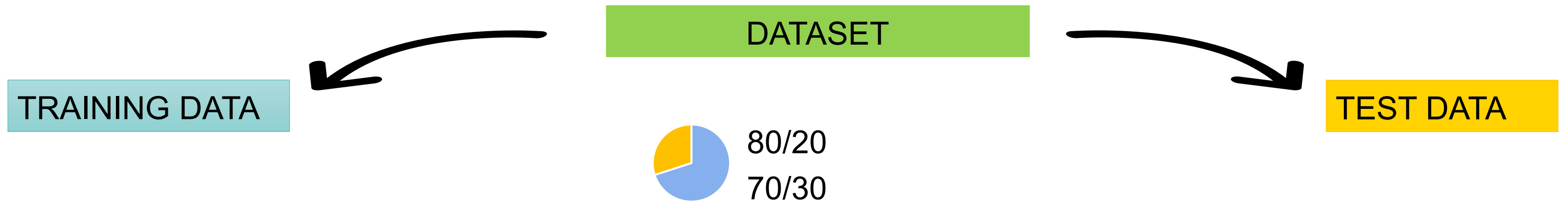


How much data? – no strict rules (cf power analysis)

- More = better
- But quality >>> quantity
- Poor data → biased or unreliable model



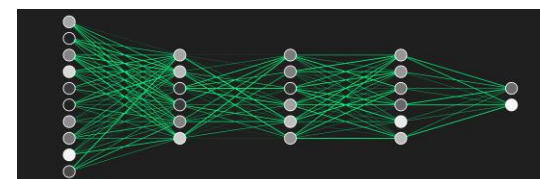
TRAIN – TEST DATA: SCENARIO 3



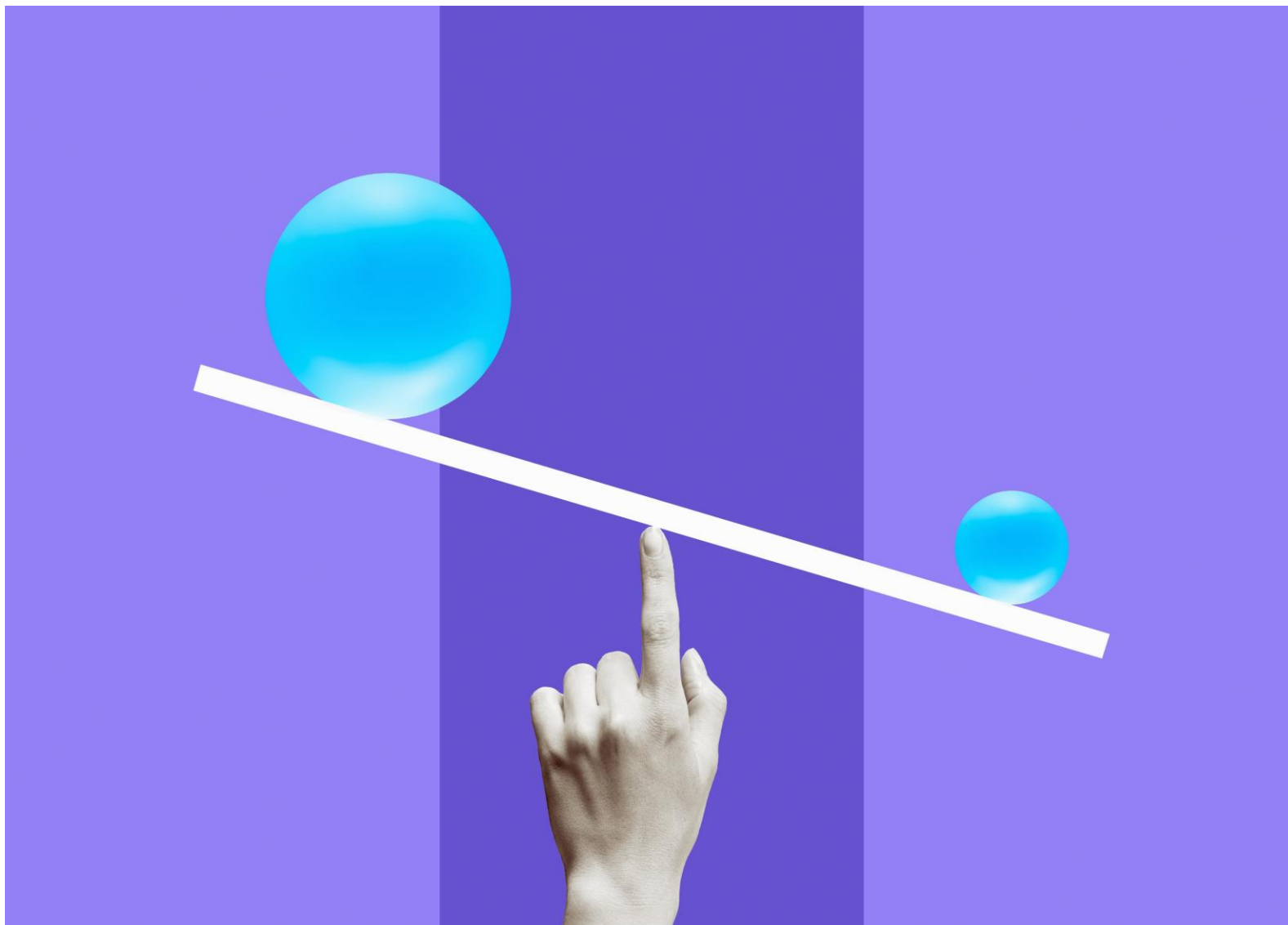
How much data? – depends on

- 1 • Model complexity

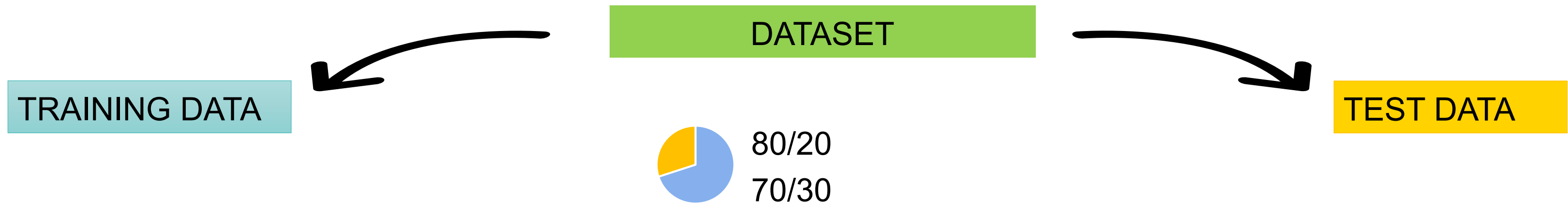
$$y = \beta_0 + \beta_1 x + \varepsilon$$



<https://towardsdatascience.com/how-to-program-a-neural-network-f28e3f38e811/>



TRAIN – TEST DATA: SCENARIO 3

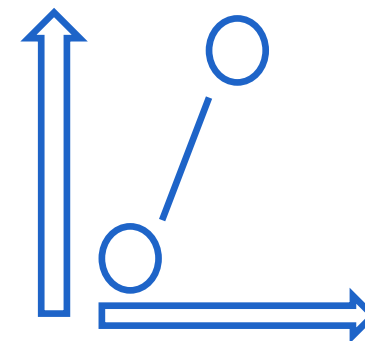


How much data? – depends on

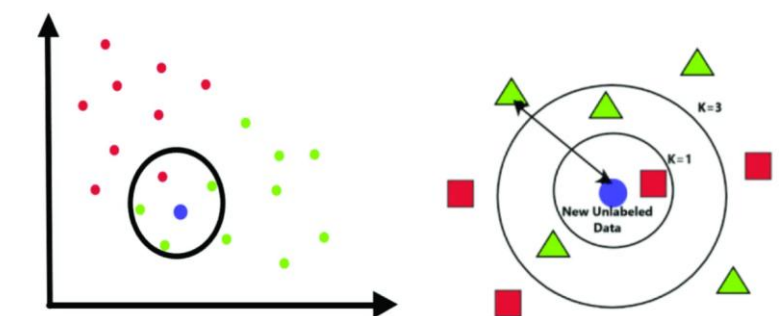
- 2 • Number of features: the curse of dimensionality



$$\text{Euclidian Distance} = \sqrt{\sum_{i=1}^n (X_1^i - X_2^i)^2}$$

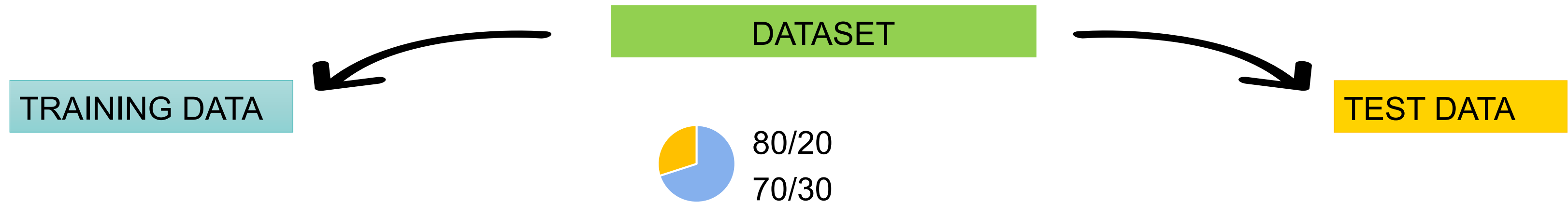


E.g. KNN



<https://ieeexplore.ieee.org/document/9065747>

TRAIN – TEST DATA: SCENARIO 3



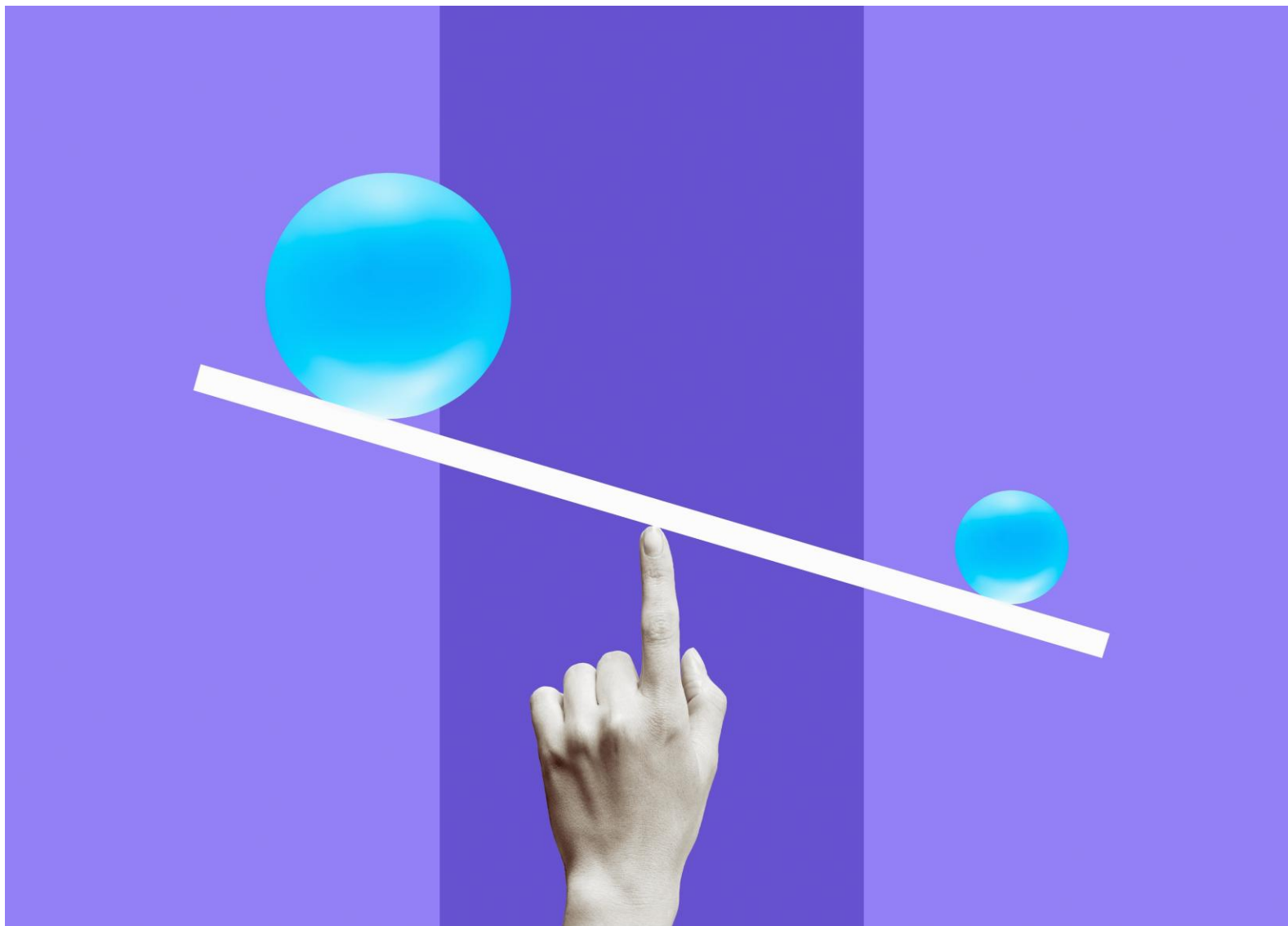
How much data? – depends on

- 3 • Task difficulty

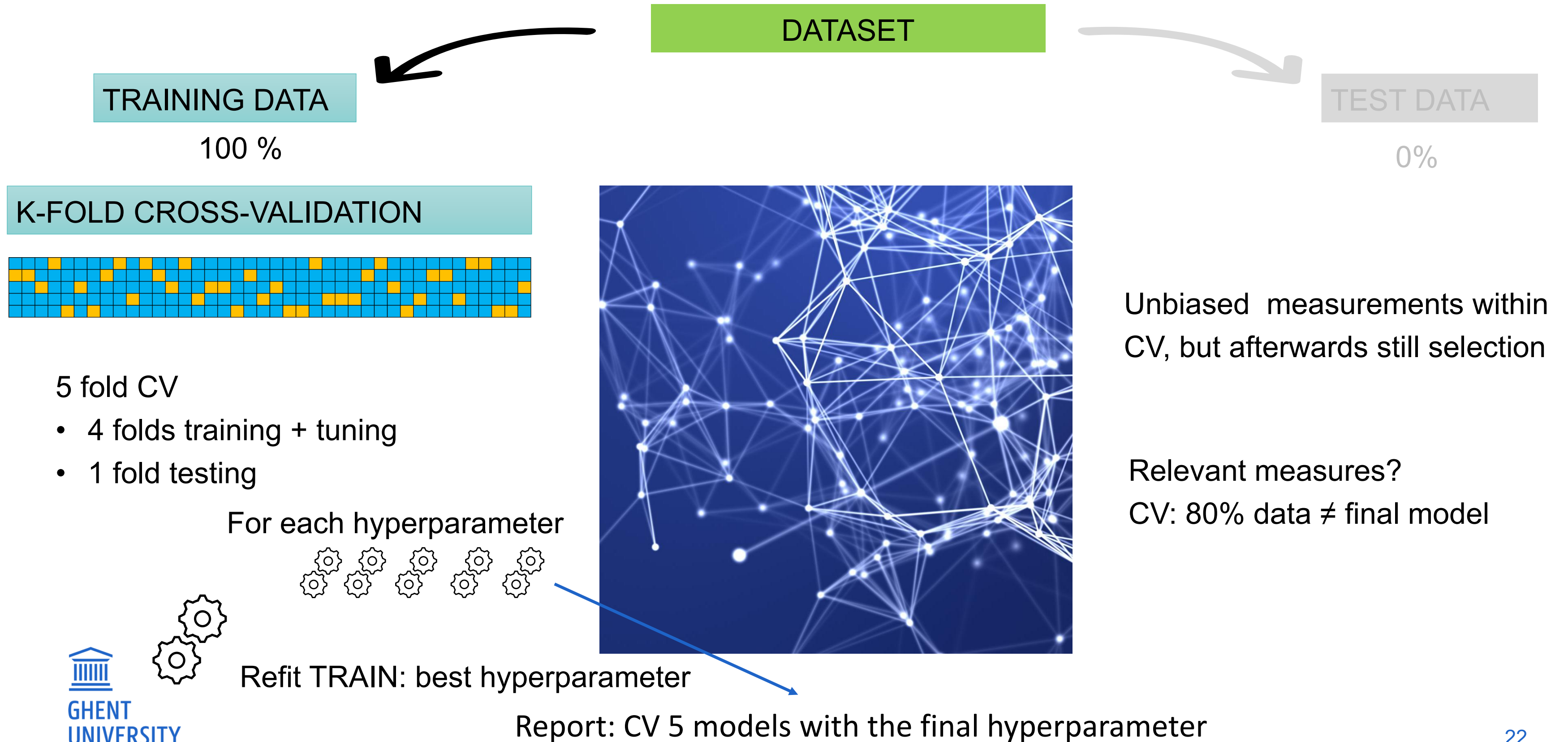
NIRS-data

Crude protein

Amino acids: high content vs low content



TRAIN – TEST DATA: SCENARIO 3



TRAIN – TEST DATA: SCENARIO ...

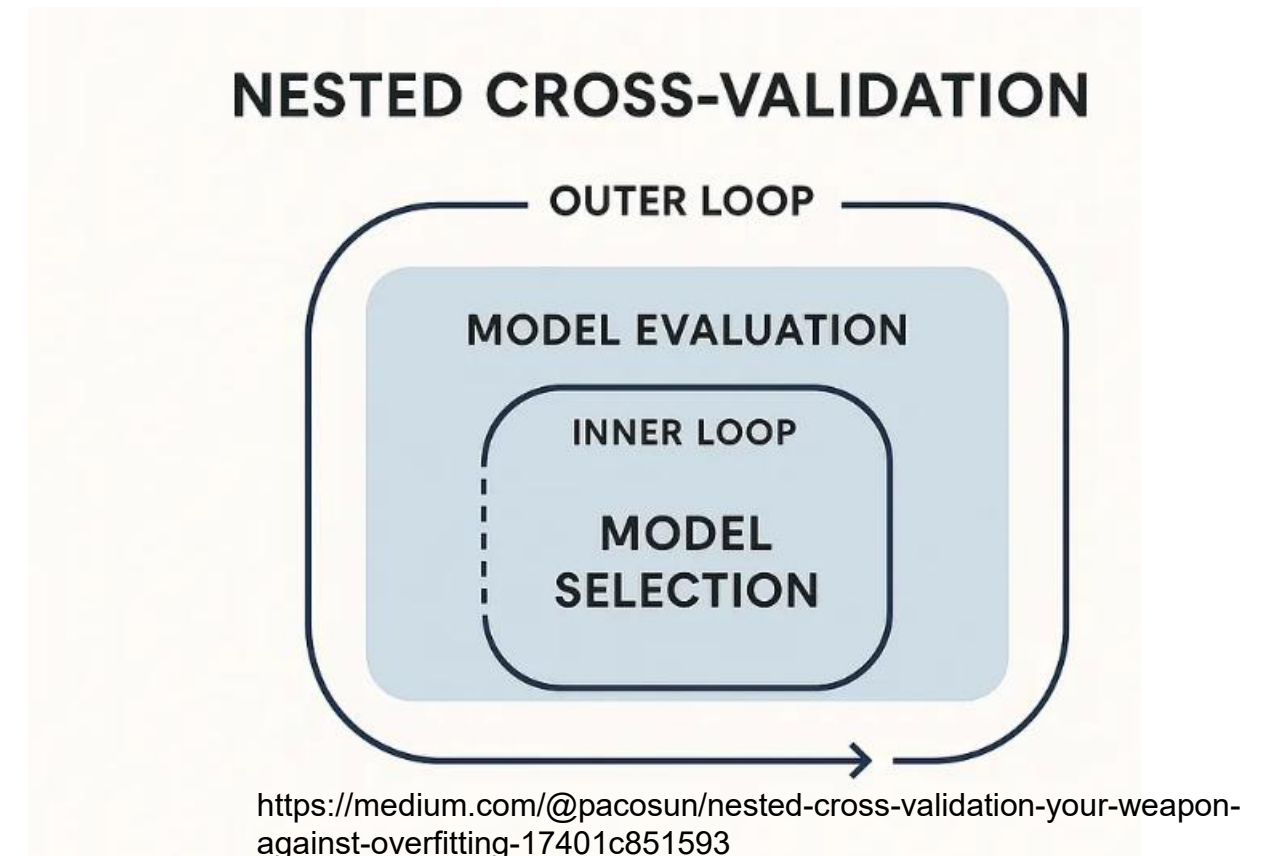
– Plenty other scenarios

Unbiased measurements within CV, but afterwards still selection
→ Nested CV

Relevant measures?

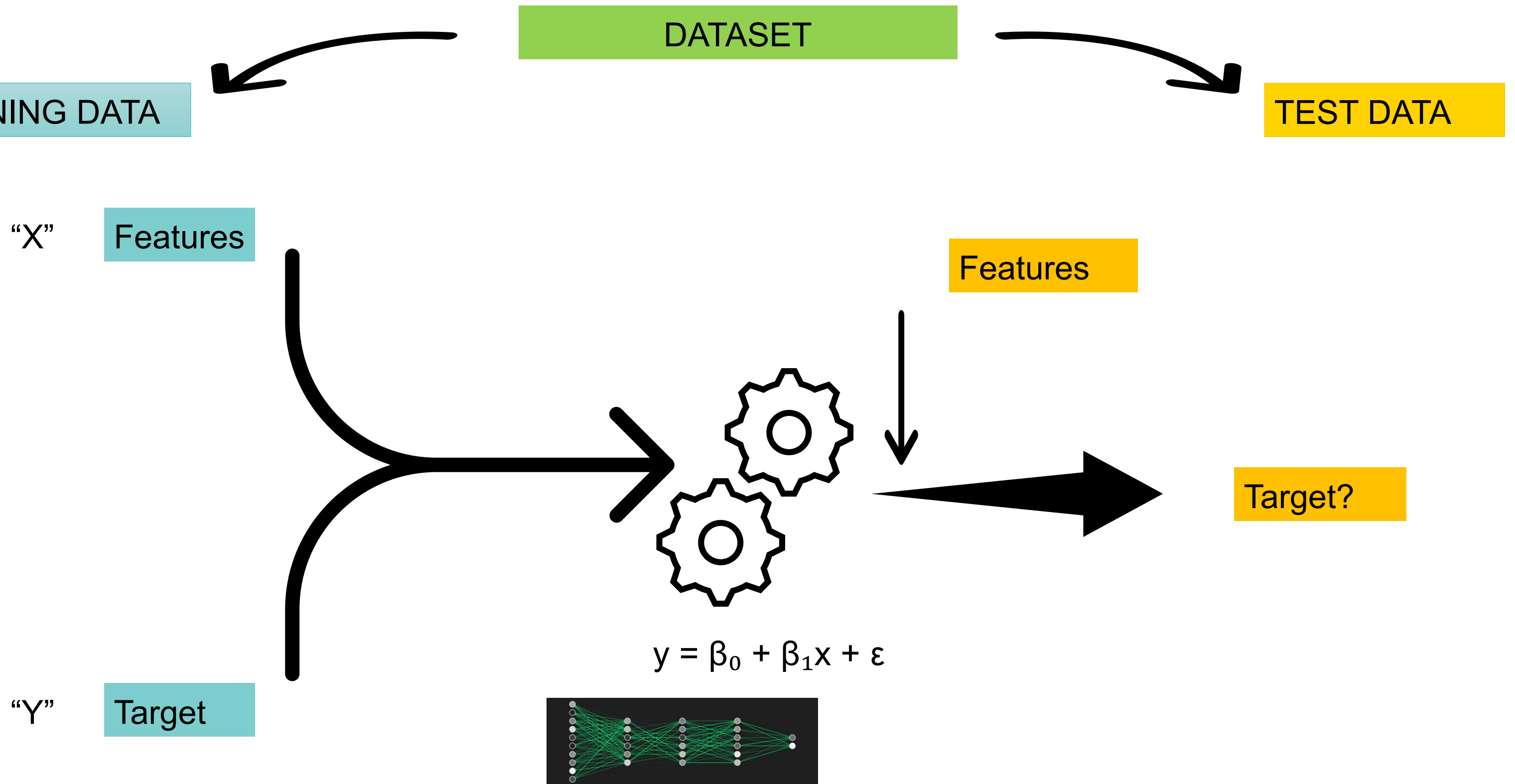
CV: 80% data $\neq$ final model

→ LOOCV



DATA LEAKAGE

CONCEPT OF SUPERVISED LEARNING



CONCEPT OF SUPERVISED LEARNING



Goal: FAIR evaluation of predictive performance

 Data leakage → failure

TYPES OF DATA LEAKAGE

1) Target data leakage

2) Train Test leakage

3) Temporal leakage

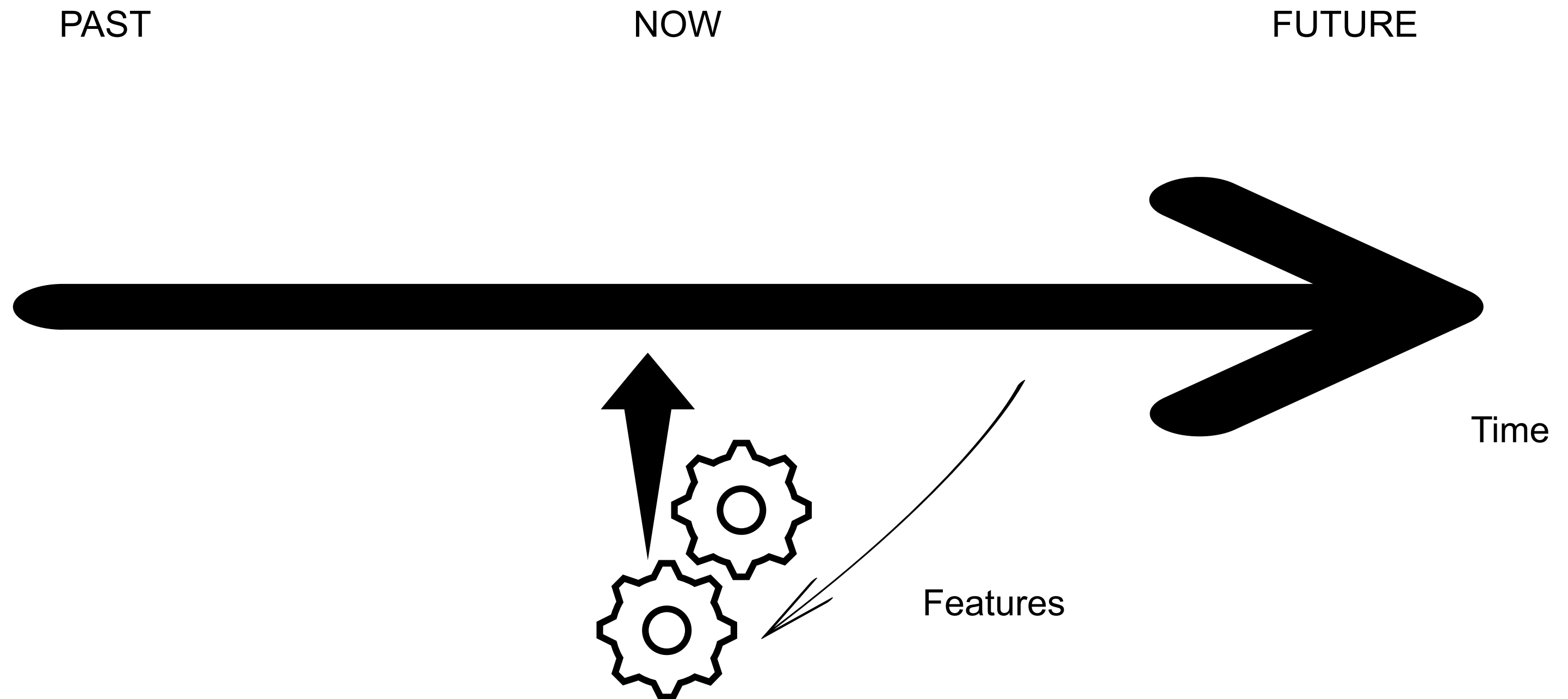
TYPES OF DATA LEAKAGE

1) Target data leakage

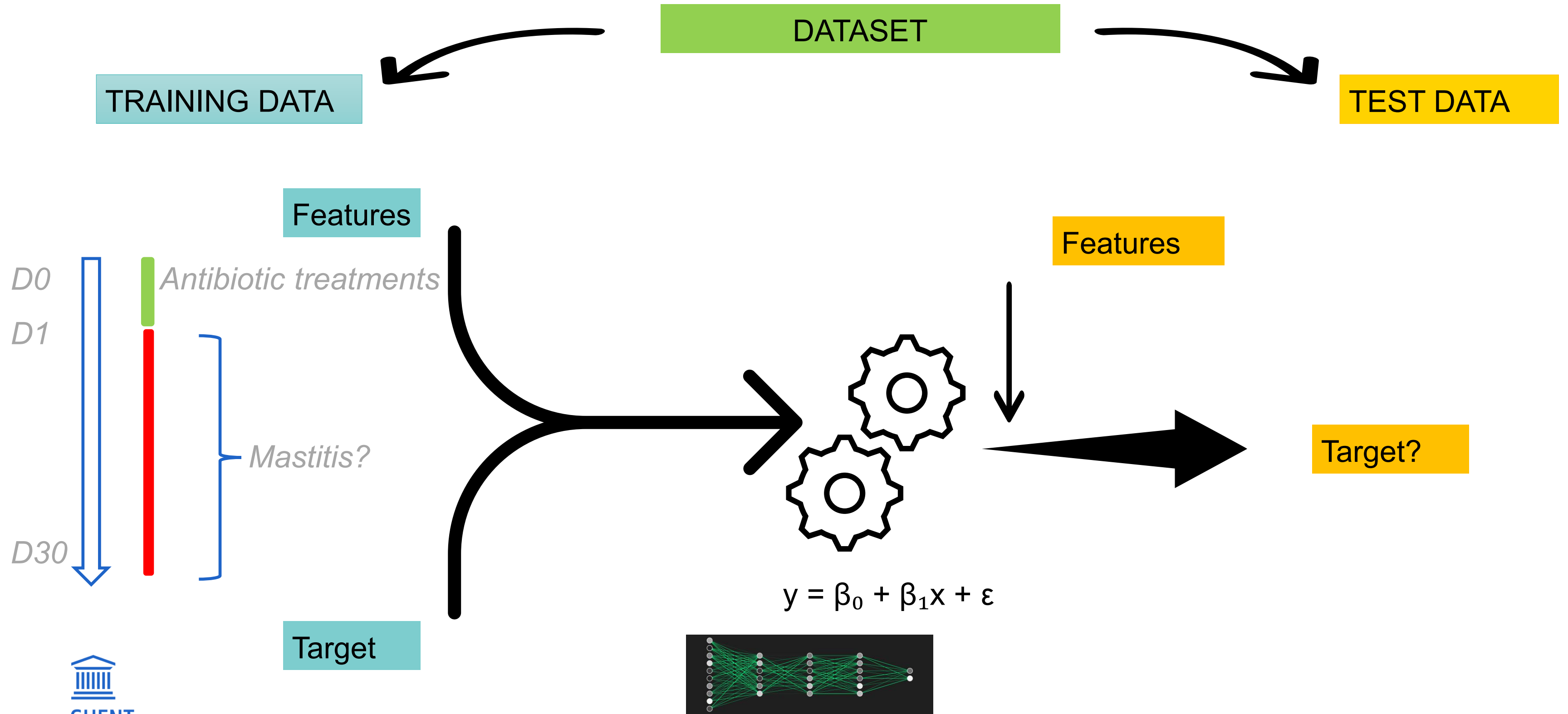
2) Train Test leakage

3) Temporal leakage

TARGET DATA LEAKAGE



TARGET DATA LEAKAGE



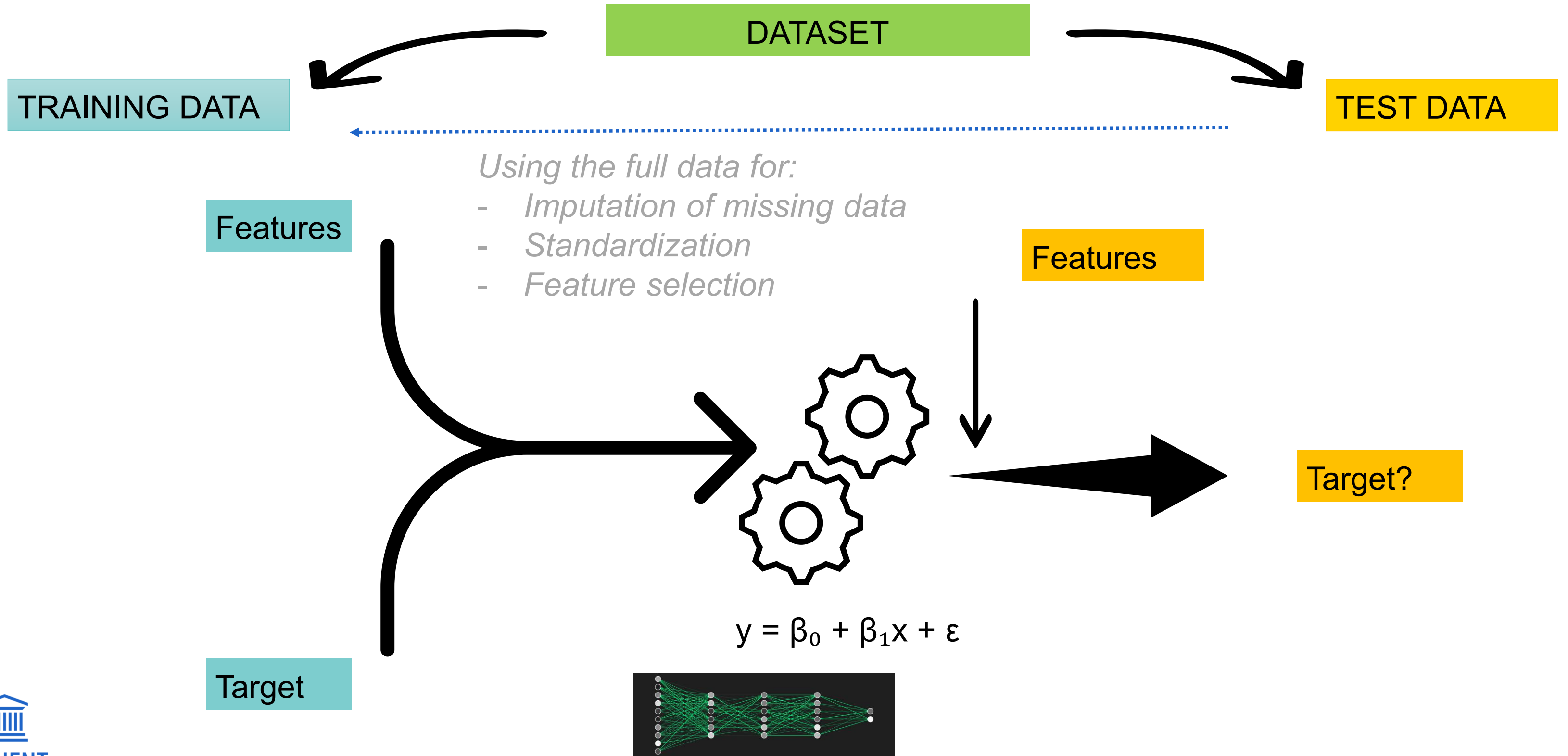
TYPES OF DATA LEAKAGE

1) Target data leakage

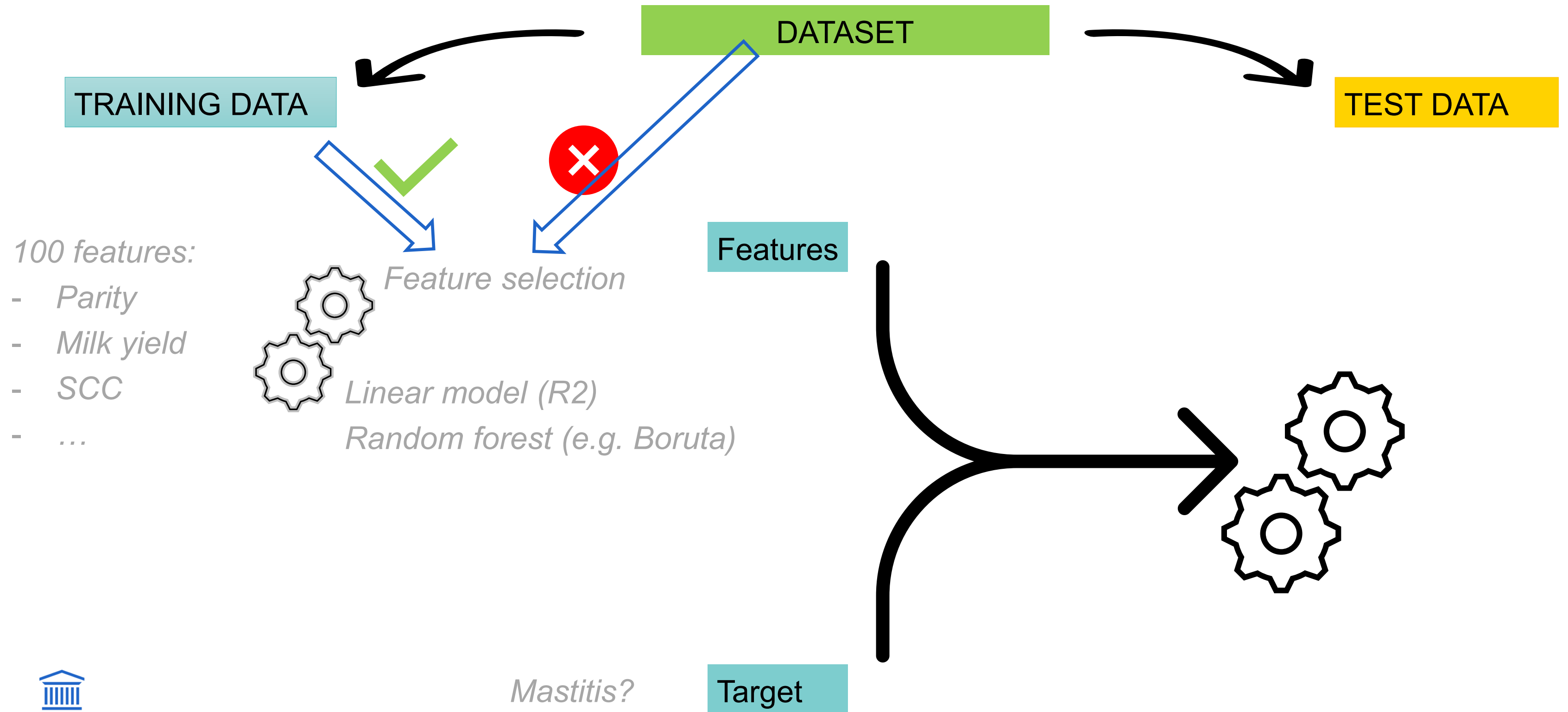
2) Train Test leakage

3) Temporal leakage

TRAIN – TEST DATA LEAKAGE



TRAIN – TEST DATA LEAKAGE



TYPES OF DATA LEAKAGE

1) Target data leakage

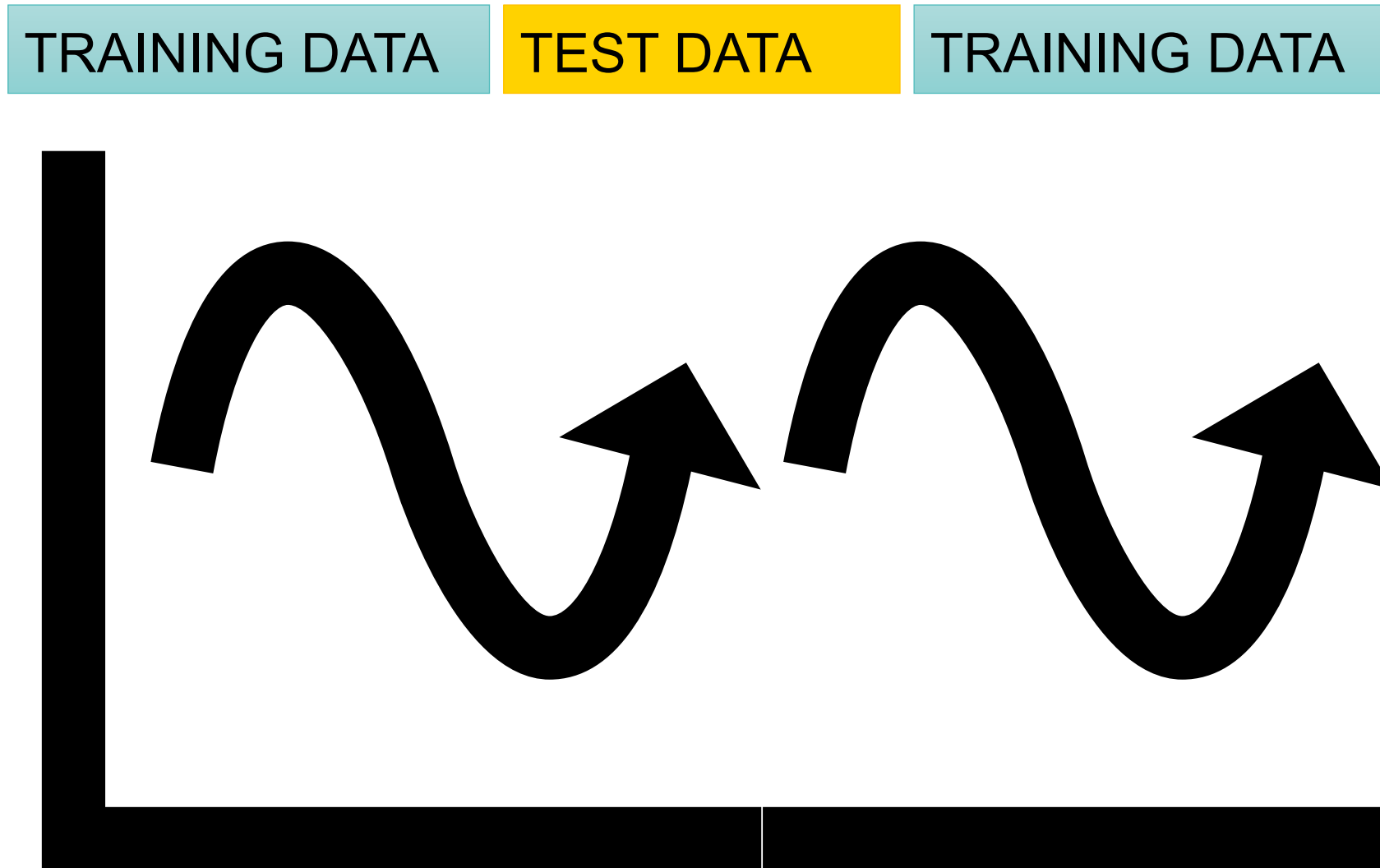
2) Train Test leakage

3) Temporal leakage

TEMPORAL LEAKAGE



*Stock price prediction:
including future data*



Timeseries data: Using future values to predict past future

Solution: only train on past timestamps to predict future

TEMPORAL LEAKAGE

Predicting future milk yield

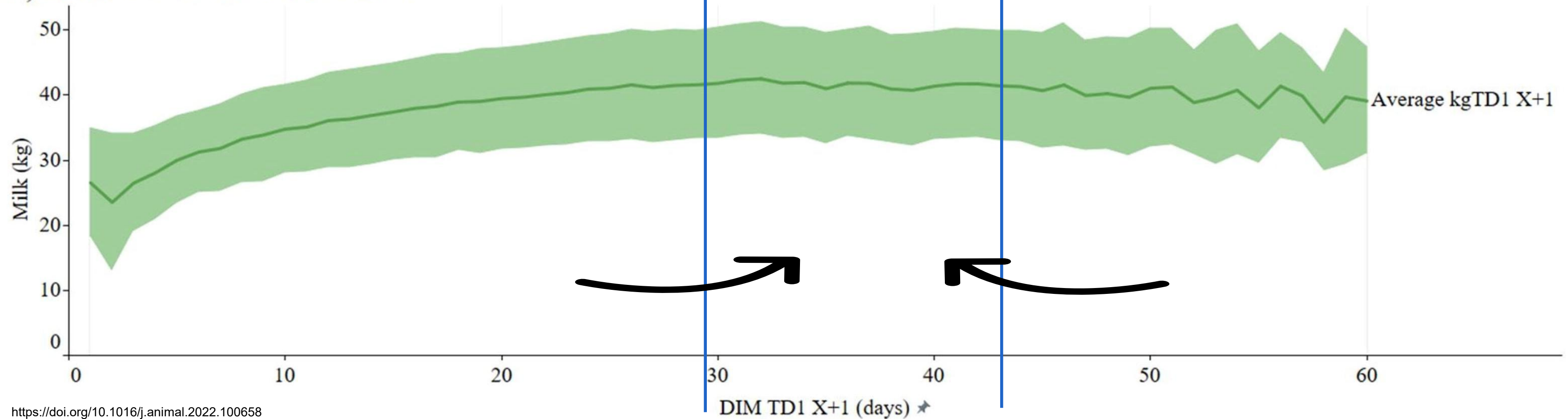


TRAINING DATA

TEST DATA

TRAINING DATA

B) Milk Production TD1 X+1



<https://doi.org/10.1016/j.animal.2022.100658>

SCENARIO 2: DATA LEAKAGE FROM IMPROPER TRAIN-TEST SPLIT

When information from the test set unintentionally influences the training process, the model appears better than it really is and fails in real-world use.



INCORRECT APPROACH

Information leaks from test data into training

All Data Together



Training and test data are mixed.

Preprocessing (eg. Scaling)



Fitted on the entire dataset (including test).

Feature Construction



Created using information from all data.

Split Into Train / Test



The test set has already influenced earlier steps.



Why this is a problem

The model gets “hints” from the test set, leading to overly optimistic performance estimates.



CORRECT APPROACH

Keep test data completely unseen until the end

Split First



Divide the data into training and test sets immediately.

Preprocess on Training Only



Fit scalers, encoders, etc. using training data only.

Feature Construction on Training



Create features using training data only.

Apply to Test & Model Training



Apply learned steps to test data, then train and evaluate.



Why this is better

The test set stays completely untouched until evaluation, giving an honest estimate of real-world performance.



Key Takeaways

- ✓ Always split your data before any preprocessing or feature engineering.
- ✓ Fit all preprocessing steps (scaling, encoding, imputation, feature selection, etc.) on training data only.
- ✓ Transform or apply those steps to the test set — never refit.
- ✓ This keeps your evaluation fair and your model trustworthy.

DATA LEAKAGE: FINAL CHECKLIST

Is My Pipeline Leaky?

- ✓ Do any features depend on the target (directly or via feature engineering)?
- ✓ Did I **split before scaling/encoding/transforming**?
- ✓ For time series: do all features use only past data?
- ✓ Would all features be available at prediction time in the real system?



<https://machinelearningmastery.com/3-subtle-ways-data-leakage-can-ruin-your-models-and-how-to-prevent-it/>



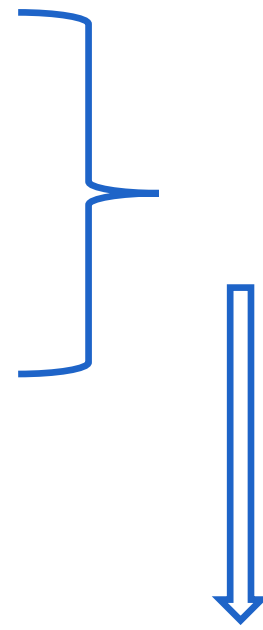
CLASS 01 EXERCISE.IPYNB:

GROUPED DATA DEPENDENCE

PITFALLS IN TRAIN-TEST SPLIT / CV

Data leakage in test data

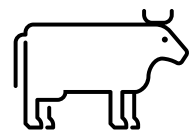
- A. Target data leakage
- B. Train Test leakage
- C. Temporal leakage



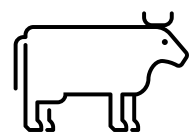
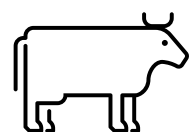
Grouped data (repeated measures)

Data from single animal: not independent
→ overoptimistic

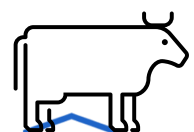
GROUPED DATA (REPEATED MEASURES)



TRAIN



TEST

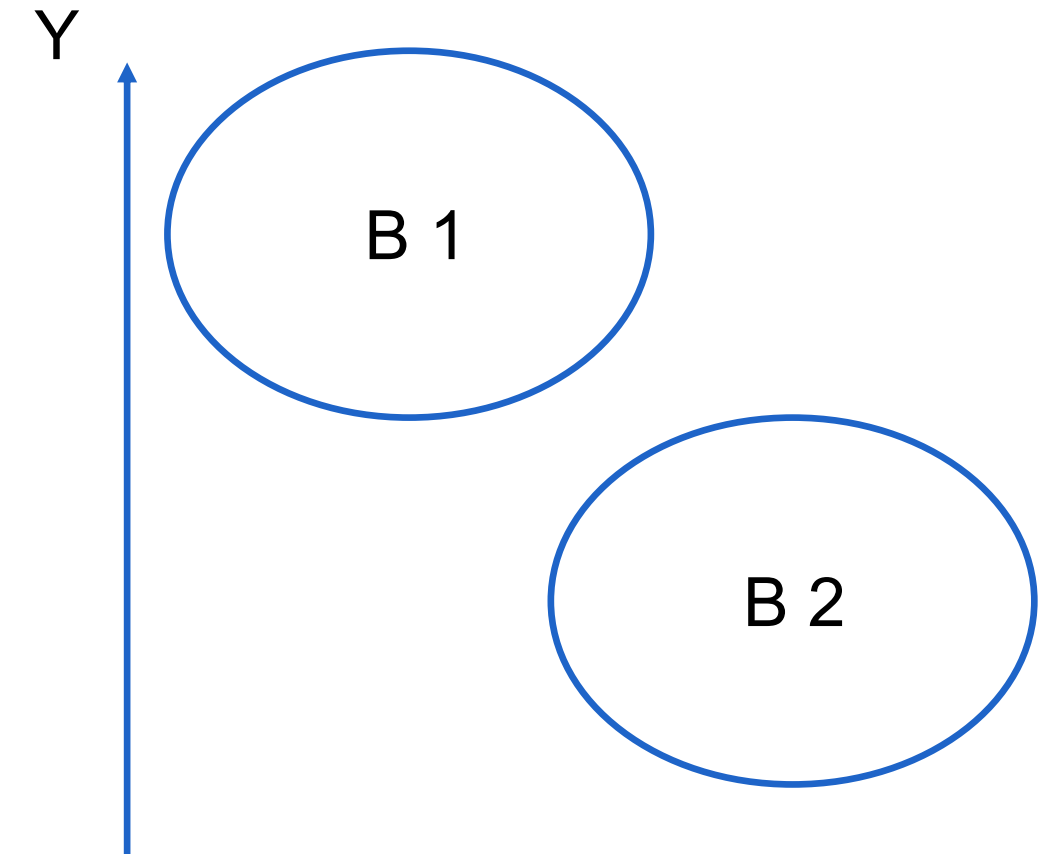


SPLITTING DATA

- No random train – test split: based on ID
- Cross-validation
 - GroupKFold
 - LGOCV

TYPICAL EXAMPLES

- Animal ID
- Farm – site
- Batch (B_i) effect in microbial data:
 - $Y_{ij} = \mu + B_i + X_j + \epsilon_{ij}$
 - Model learns batch
 - Detection: e.g. PCA, Permanova
 - Solution: batch-correction methods
→ TRAIN



DATA IMBALANCE

CLASSIFICATION IN ANIMAL SCIENCE



Research article • Open access Under a Creative Commons license ↗ Get rights and content ↗

The diversity in dairy cattle reticulorumen temperature: Identifying water intake events

A.K. Shirley ^a  , P.C. Thomson ^b, A. Chlingaryan ^a, C.E.F. Clark ^c [Show more](#) ▼

[View PDF](#) [Download full issue](#) [Cite](#) [Add to Mendeley](#) [Share](#) | [10.1016/j.compag.2025.110357](https://doi.org/10.1016/j.compag.2025.110357) ↗

Article | [Open access](#) | Published: 06 October 2025

Improving bovine disease detection through multilabel classification

[Ghalib Nadeem](#), [Muhammad Fahim Ul Haque](#), [Hameeza Ahmed](#), [Mansoor Ebrahim](#), [Sadique Ahmad](#), [Isabel de la Torre Díez](#), [Hanaa A. Abdallah](#) & [Abdelhamied A. Ateya](#) 

[Scientific Reports](#) **15**, Article number: 34766 (2025) | [Cite this article](#)

Accurate prediction of calving in dairy cows by applying feature engineering and machine learning

[Jorge A. Vázquez-Diosdado](#) ^{a 1}  , [Julien Gruhier](#) ^{b 1}, [G.G. Miguel-Pacheco](#) ^{a c}, [Martin Green](#) ^a, [Tania Dottorini](#) ^a, [Jasmeet Kaler](#) ^a

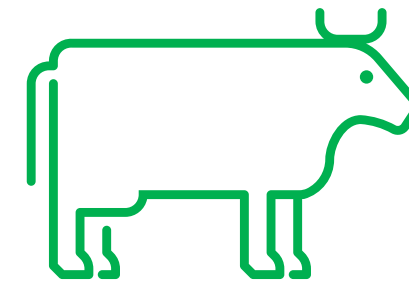
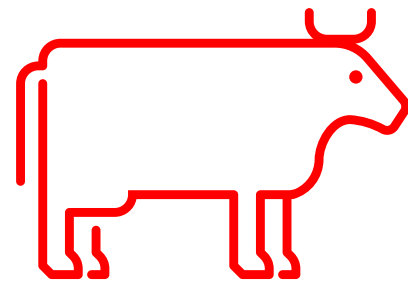
[Show more](#) ▼
[Add to Mendeley](#) [Share](#) [Cite](#)
<https://doi.org/10.1016/j.prevetmed.2023.106007> ↗

[Get rights and content](#) ↗

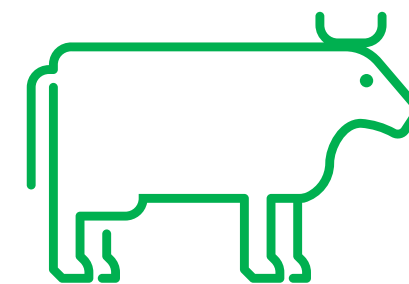
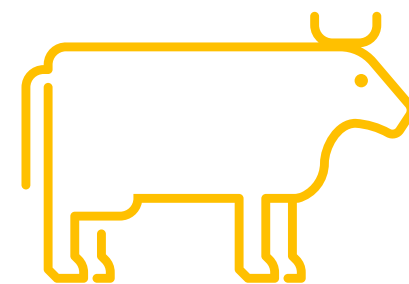
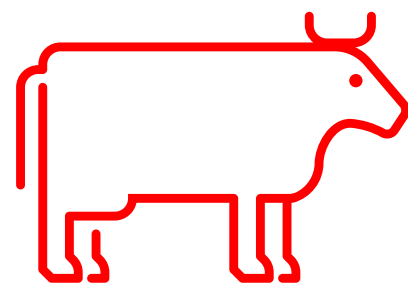
Diagnostic milk biomarkers for predicting the metabolic health status of dairy cattle during early lactation

[S. Heirbaut](#) ¹ · [X.P. Jing](#) ^{1,2} · [B. Stefańska](#) ³ · ... · [M. Thys](#) ⁷ · [L. Vandaele](#) ⁷ · [V. Fievez](#) ¹   ... [Show more](#)

CLASSIFICATION

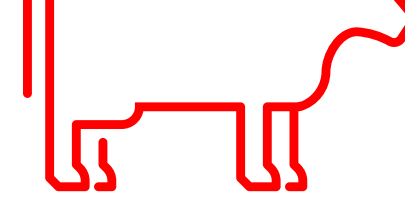


2 classes



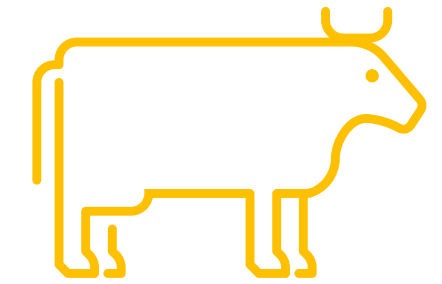
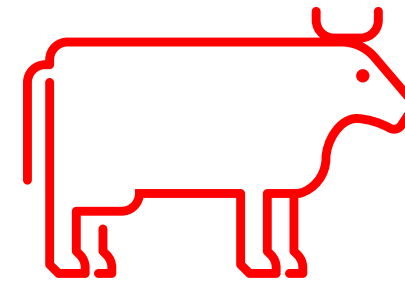
Multiclass

CLASSIFICATION

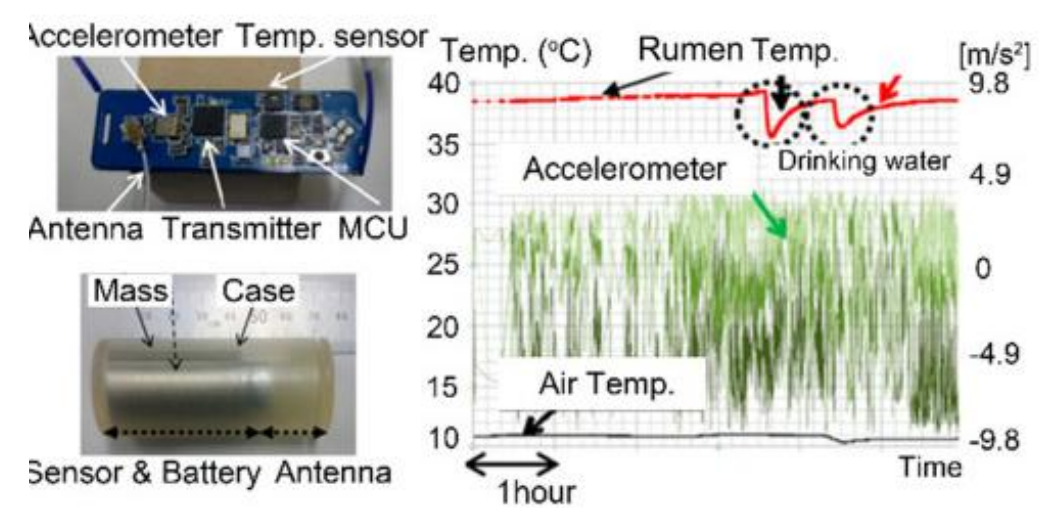


Evaluation of model performance?

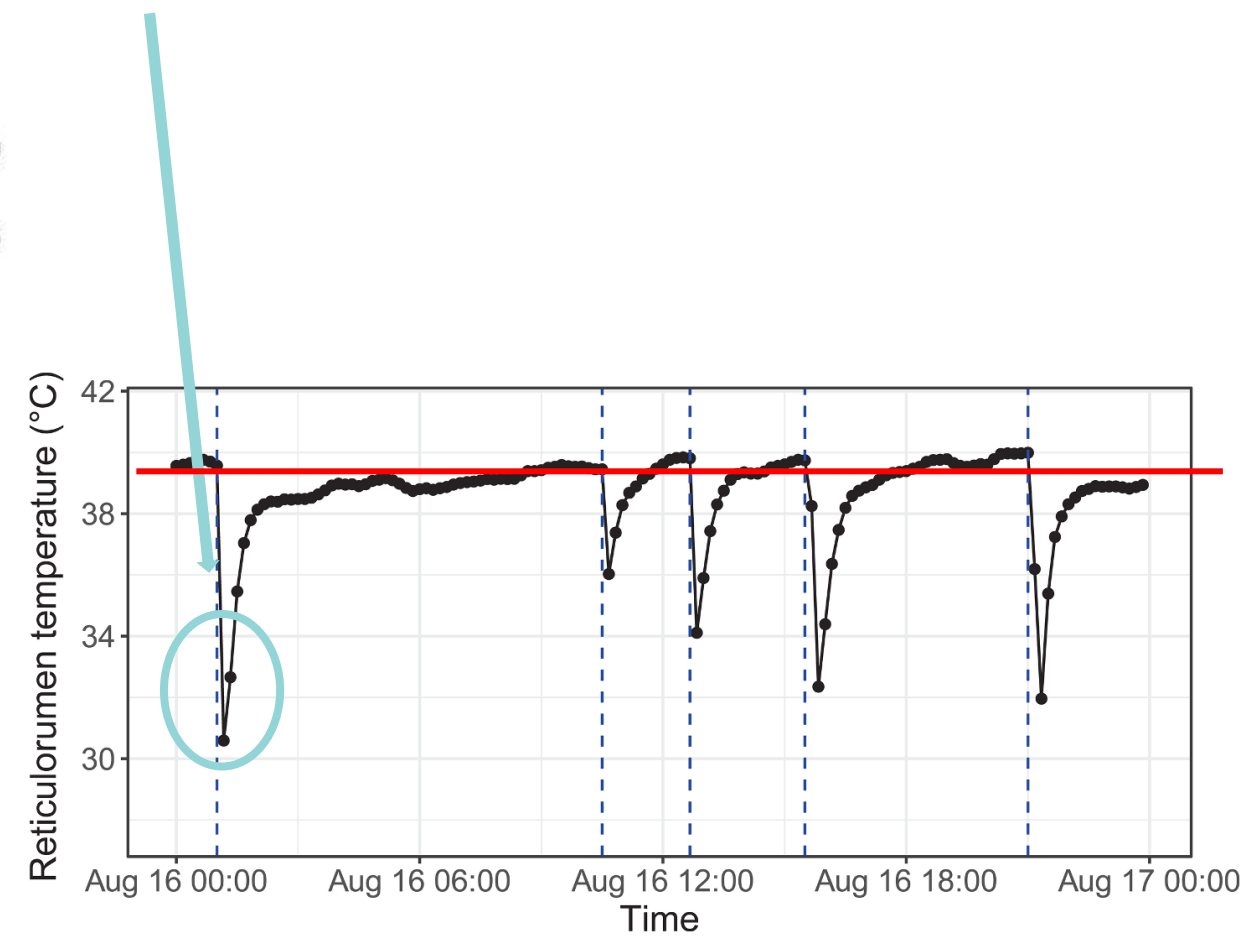
- Accuracy
- Precision
- Recall (sensitivity)
- Specificity
- F1-score
- AUC ROC-curve
- AUC PR-curve



CLASSIFICATION METRICS: AN EXAMPLE



Su Han et al. (2022)
<https://doi.org/10.3168/jds.2021-20576>



<https://doi.org/10.1016/j.compag.2025.110357>

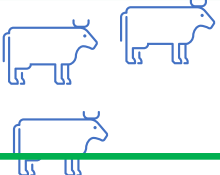
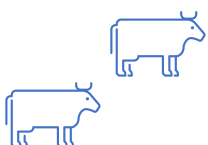
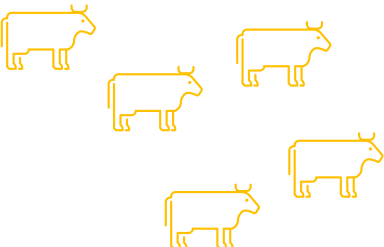
CLASSIFICATION METRICS: AN EXAMPLE

$$\text{Precision} = \frac{TP}{TP+FP}$$

$$\text{Recall} = \frac{TP}{TP+FN}$$

$$\text{F-score} = \frac{2 \cdot \text{Pre} \cdot \text{Rec}}{\text{Pre} + \text{Rec}}$$

PREDICTION: Drinking?

		OBSERVATION: Drinking?	
		YES	NO
PREDICTION: Drinking?	YES	TP 	FP 
	NO	FN 	TN 

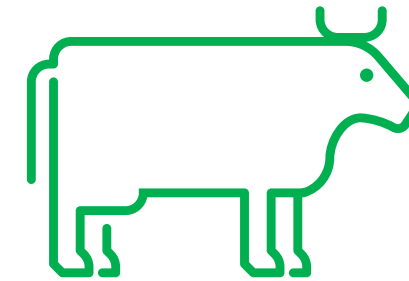
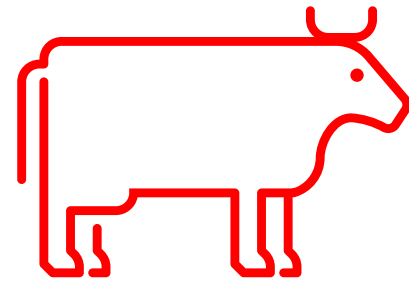
$$\text{Precision} = \frac{3}{3+1} = 75\%$$

$$\text{Recall} = \frac{3}{3+2} = 60\%$$

CLASSIFICATION METRICS: AN EXAMPLE

Most important?

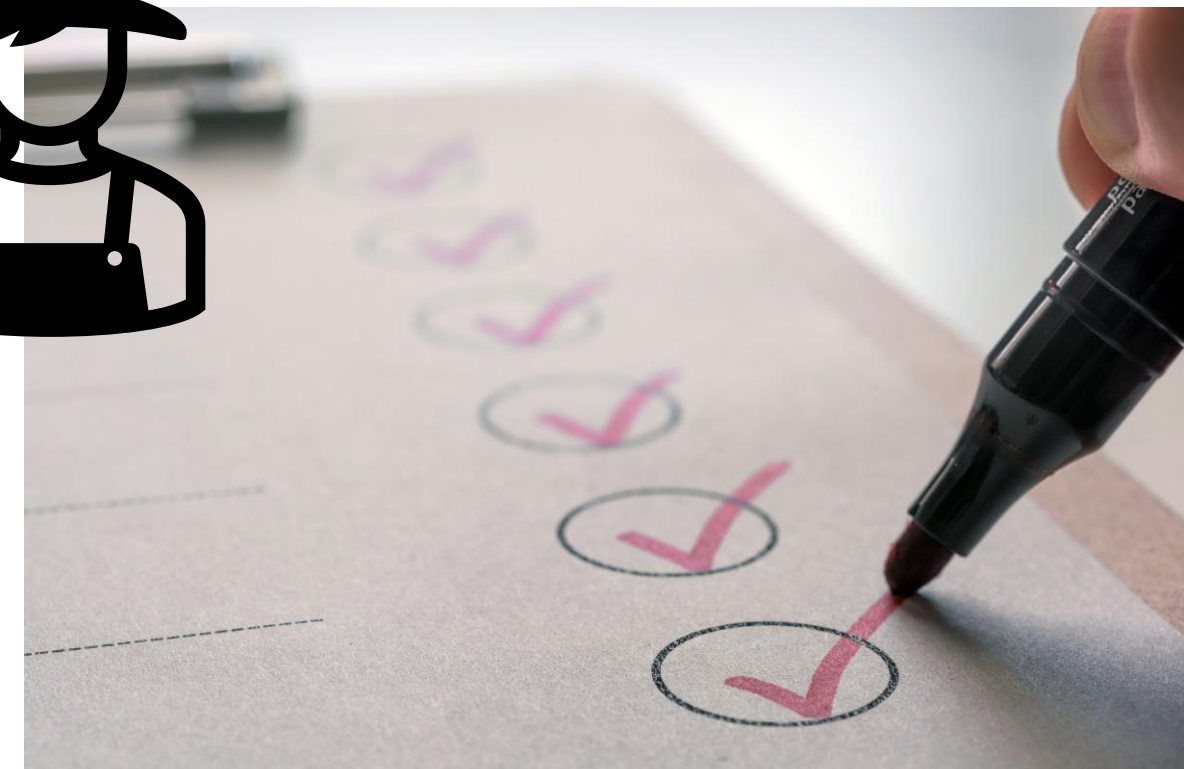
$$\text{Recall} = \frac{TP}{TP+FN}$$



Is it okay to “miss” some animals?

$$\text{Precision} = \frac{TP}{TP+FP}$$

How many FP are “acceptable”?



CLASSIFICATION: ACCURACY

So why do we need all the other metrics?

$$\text{Accuracy} = \frac{TP+TN}{TP+FP+FN+TN}$$

PREDICTION: Drinking?

$$\text{Accuracy} = \frac{3+5}{3+5+2+1} = 73\%$$

→
$$\text{Accuracy} = \frac{3+20}{3+20+2+1} = 88\%$$

$$\text{Recall} = \frac{3}{3+2} = 60\%$$

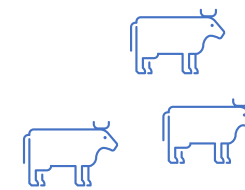
OBSERVATION: Drinking?

YES

NO

YES

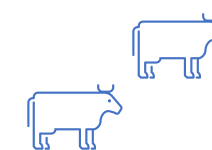
TP



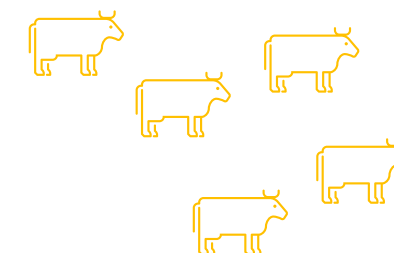
FP



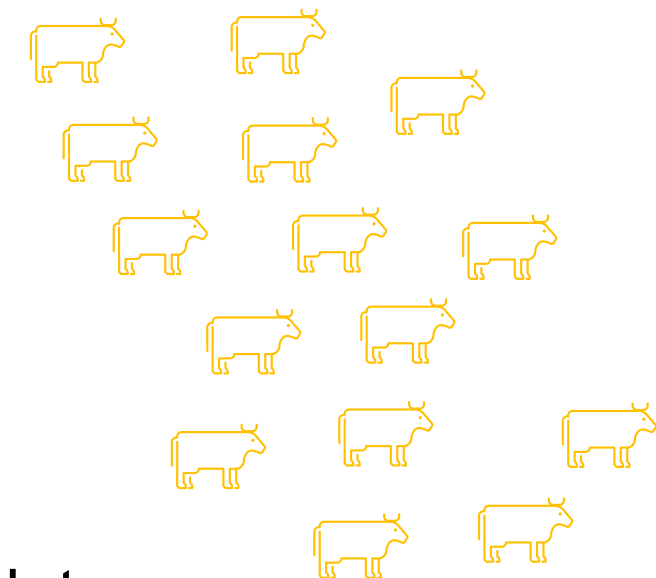
FN



TN



$$\text{Precision} = \frac{3}{3+1} = 75\%$$



Real life: often imbalanced data

Chance drinking event over 24h?

CLASSIFICATION: ACCURACY

Often: data = more imbalanced

Now 5/31 drinking events

If 5 min drinking/day

Just predicting majority class

5/1440 → accuracy

$$\text{Accuracy} = \frac{TP+TN}{TP+FP+FN+TN} = \frac{0+1435}{0+0+5+1435} = 99.7\%$$

DATA IMBALANCE: SOLUTION

- Making it balanced?
 - Downsampling
 - Upsampling
- Always on training data
- If not:
 - Test data is not representative
 - Data leakage

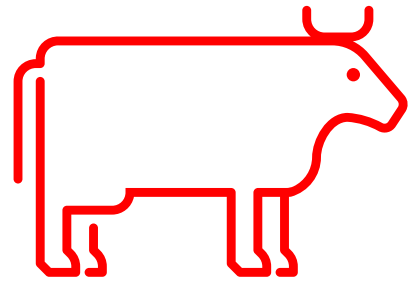
DATA IMBALANCE: SOLUTION

- Making it balanced?
 - Downsampling
 - Upsampling
- How?
 - Remove/duplicate training instances
 - SMOTE (Synthetic Minority Over-sampling Technique)
 - ...

INTERMEZZO SMOTE

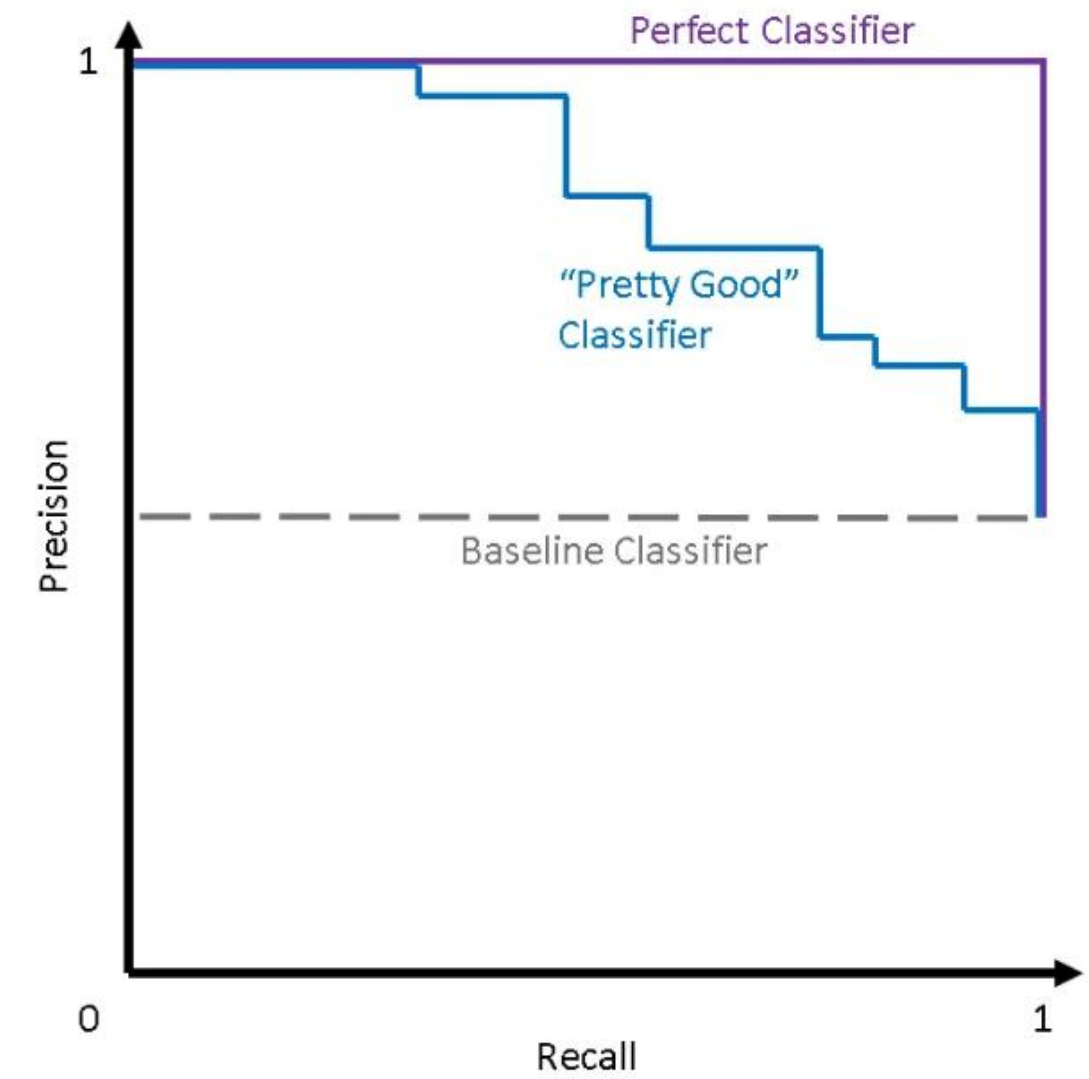
- Creates synthetic samples of the minority class
- Selecting random minority class sample
- Finding its k nearest neighbors
- Randomly selecting one of the k neighbours
- Creating new sample: linear combination original sample and selected neighbour

PRECISION-RECALL CURVE



Probability (0 – 1)

If > 0.5 : P



But not necessarily the best cut-off

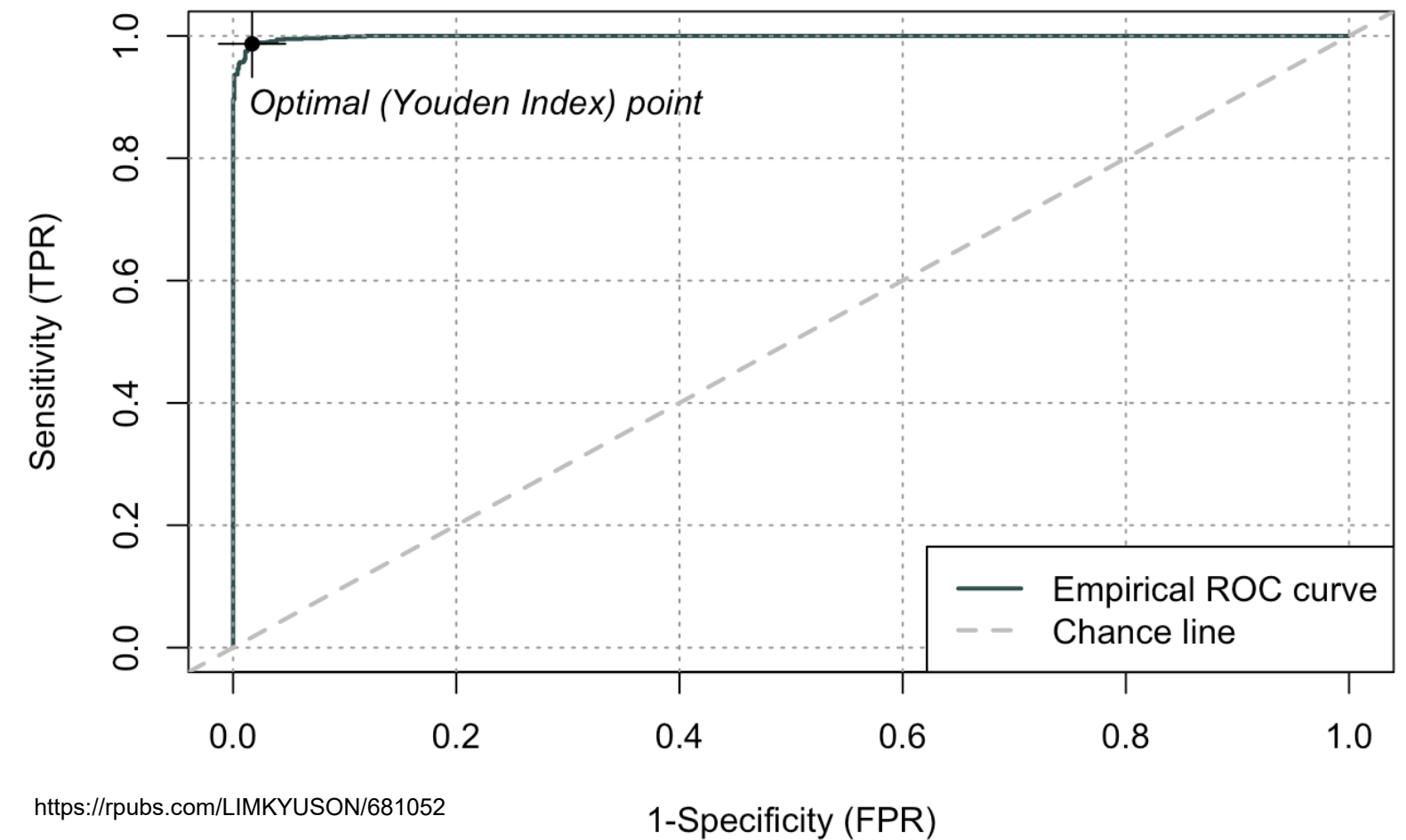
How to optimize?

PRECISION-RECALL

- How to optimize?
- F1-score
- Youden index (J)
 $J = \text{sensitivity} + \text{specificity} - 1$

Select cut-off (i) for $J_{\max}$

Important: i = tuning process → training data



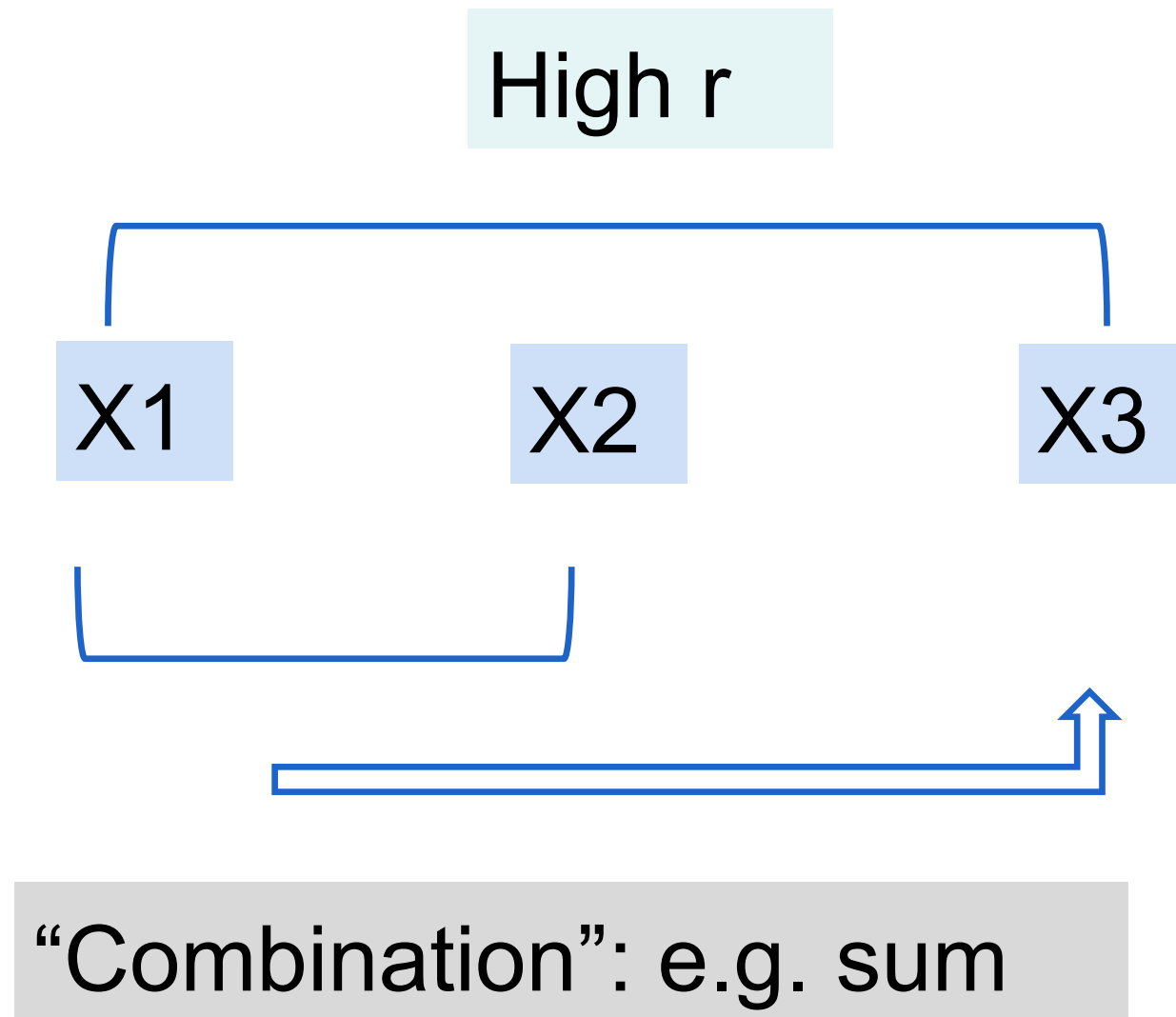


CLASS 02 EXERCISE.IPYNB

MULTICOLLINEARITY

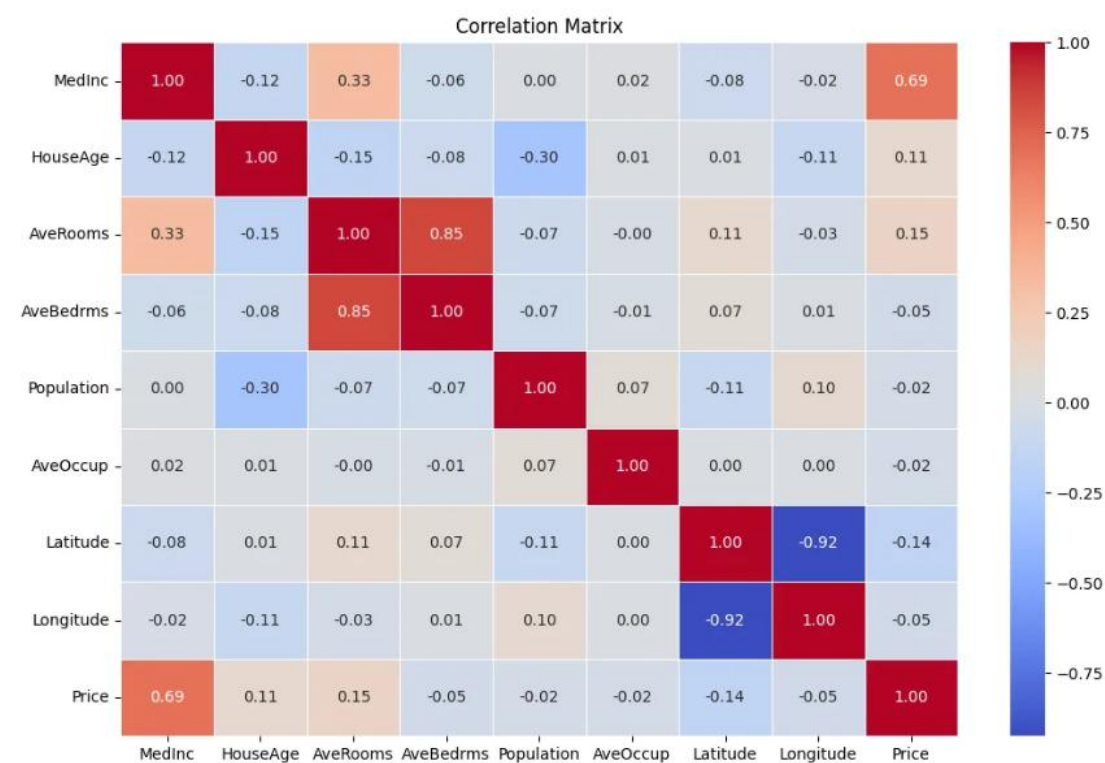
DEFINITION

- Predictors in a regression model are linearly dependent

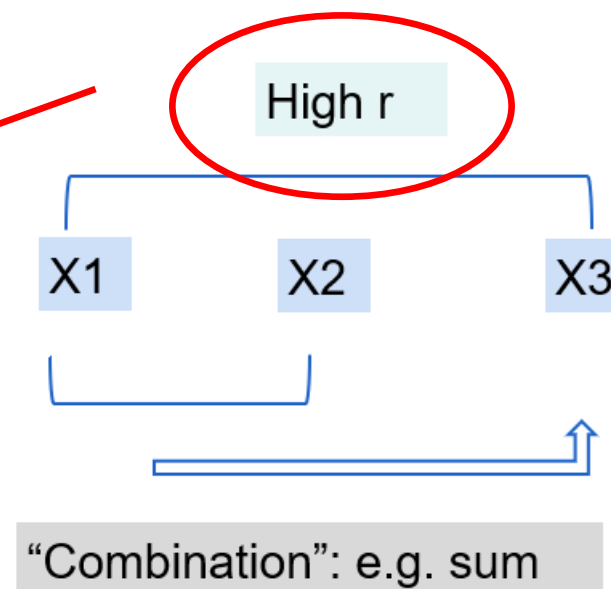


Feature engineering X3
Keep X1 – X2?

HOW TO DETECT?



Source: <https://www.geeksforgeeks.org/house-price-prediction-with-catboost/>



Not “visible” in
correlation matrix

Variance inflation factor (VIF)

$$VIF_i = \frac{1}{1 - R_i^2}$$

VIF_{X3}

$X3 \sim X1 + X2$

Eigenvalues, rank of matrix, etc.

A PROBLEM?

It depends...

Statistics

Coefficients: unreliable

Predictive modeling

Predicting well?

Model interpretation

Complex, computation

SOLUTIONS

- Remove correlated variables
- Combining correlated variables: feature engineering
- Dimension reduction
- Feature selection
 - Domain knowledge
 - Model based and/or shrinkage methods

RIDGE AND LASSO REGRESSION

– Least squares: coefficients $\rightarrow$ min RSS
$$\text{RSS} = \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2.$$

– Ridge regression
$$\sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p \beta_j^2 = \text{RSS} + \lambda \sum_{j=1}^p \beta_j^2,$$

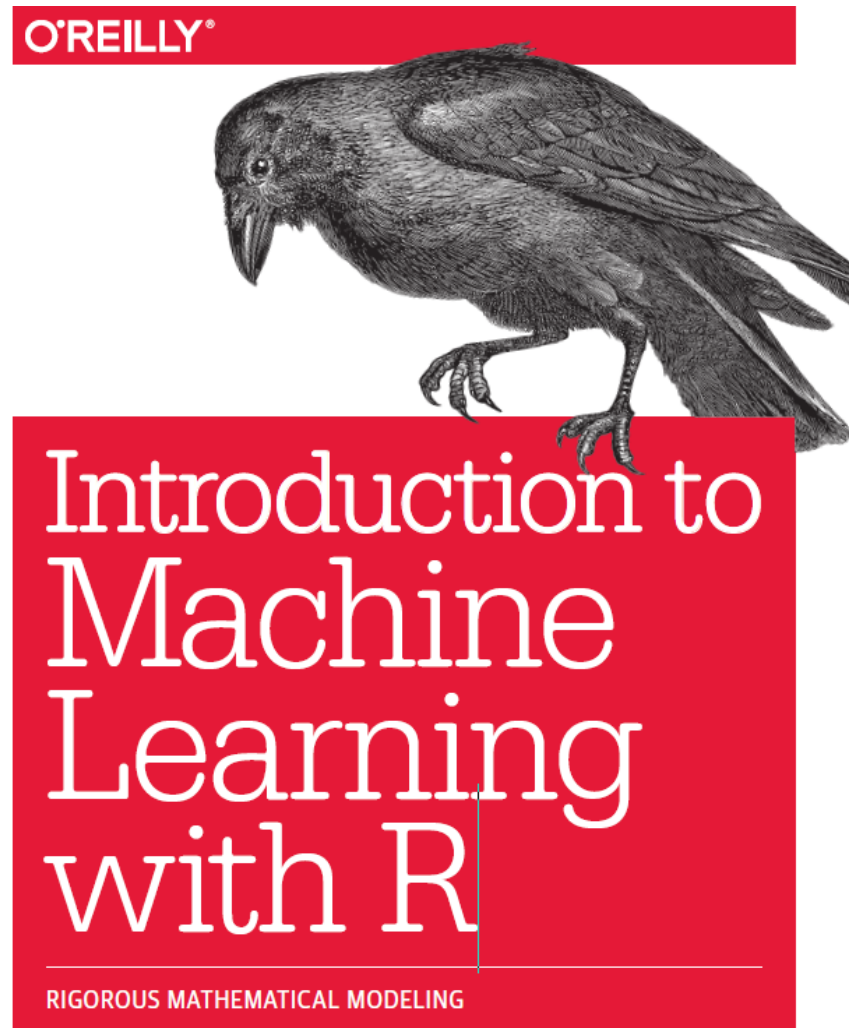
– Lasso regression
$$\sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p |\beta_j| = \text{RSS} + \lambda \sum_{j=1}^p |\beta_j|.$$

Feature selection: always on train data



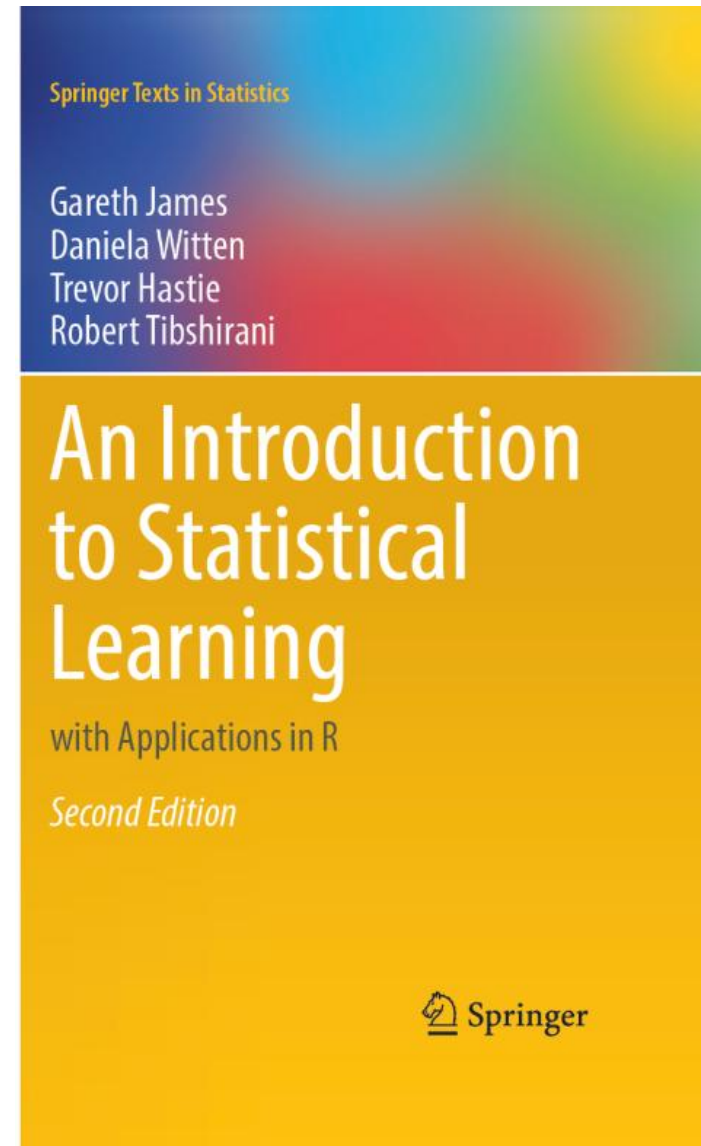
CLASS 03 EXERCISE.IPYNB

REFERENCES

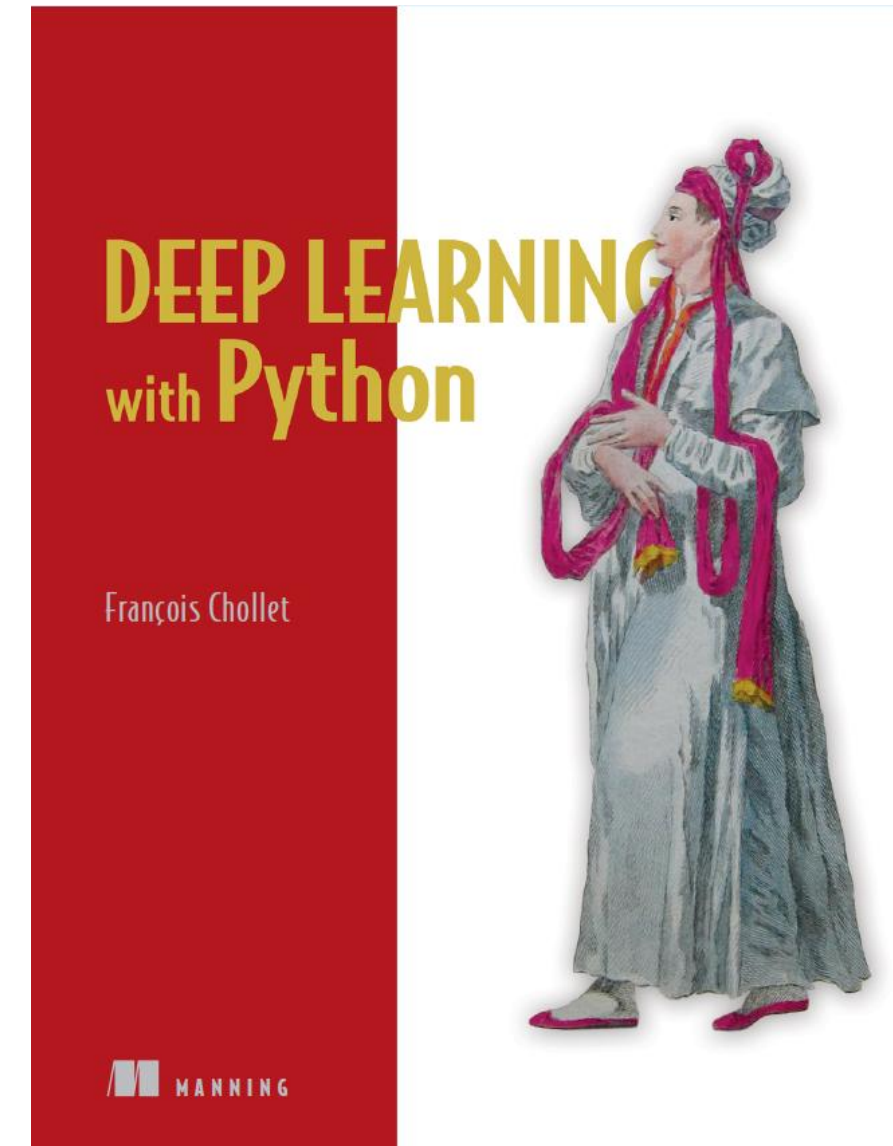
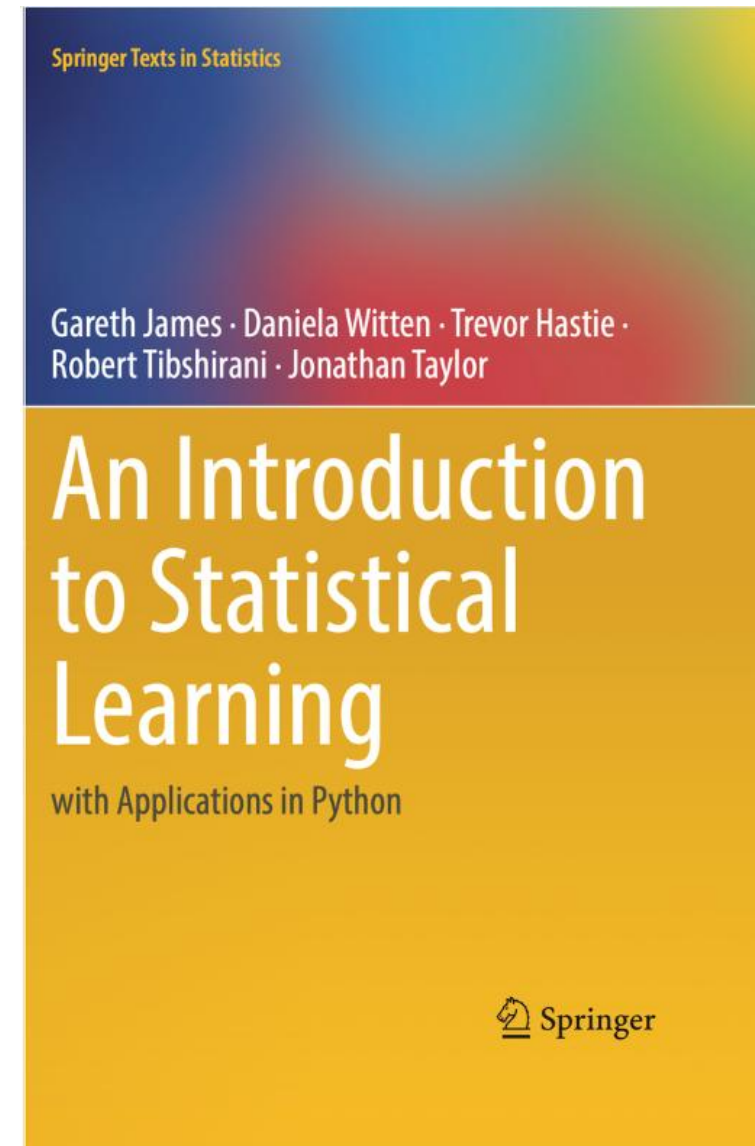


Scott V. Burger

<https://www.oreilly.com/library/view/introduction-to-machine/9781491976432/>



<https://www.statlearning.com/>



<https://deeplearningwithpython.io/>

Department of Animal Sciences and Aquatic
Ecology

E
jeroen.degroote@ugent.be
stijn.heirbaut@ugent.be

www.ugent.be

 Universiteit Gent
 @ugent
 @ugent
 Ghent University